

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

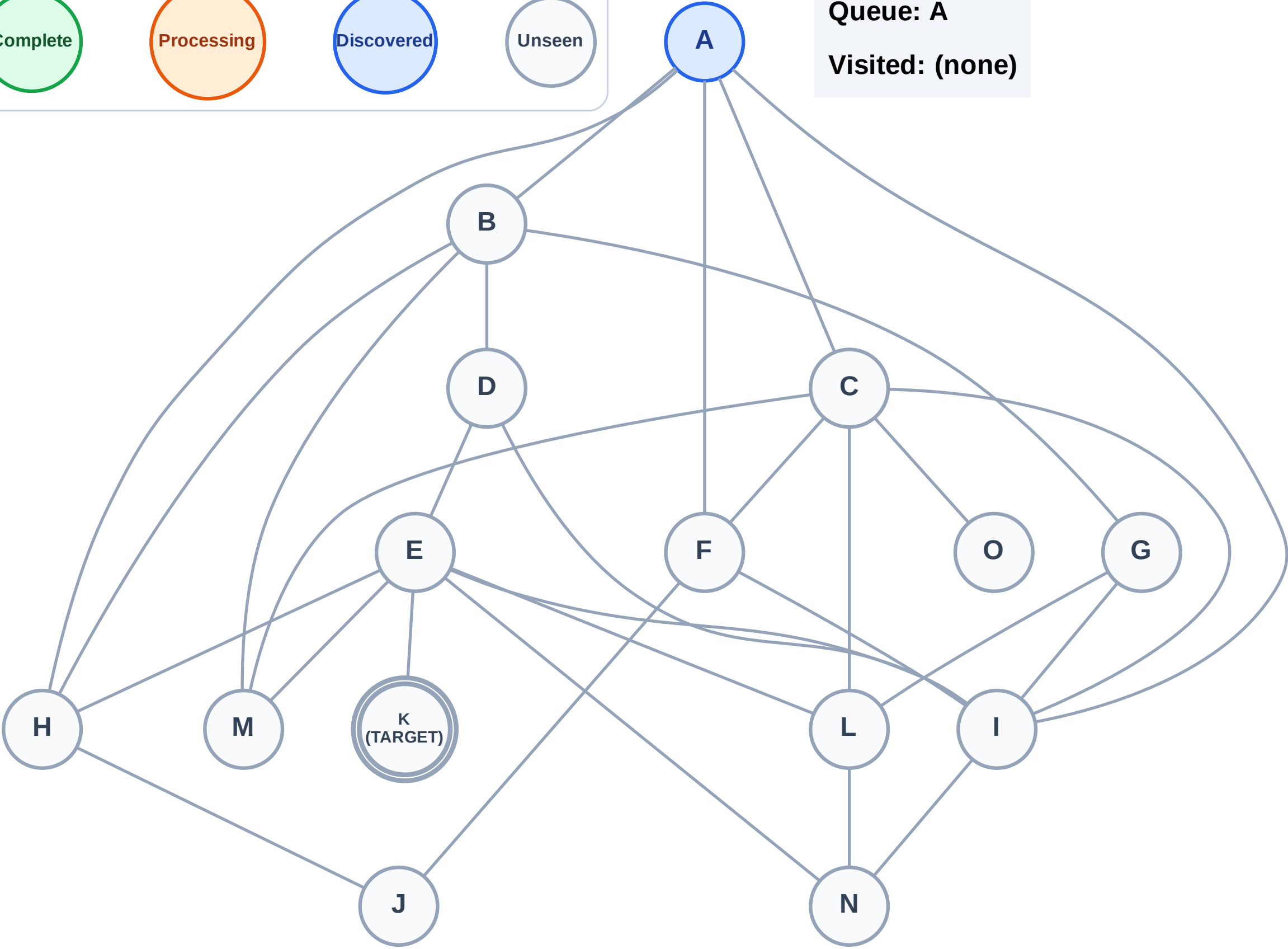
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

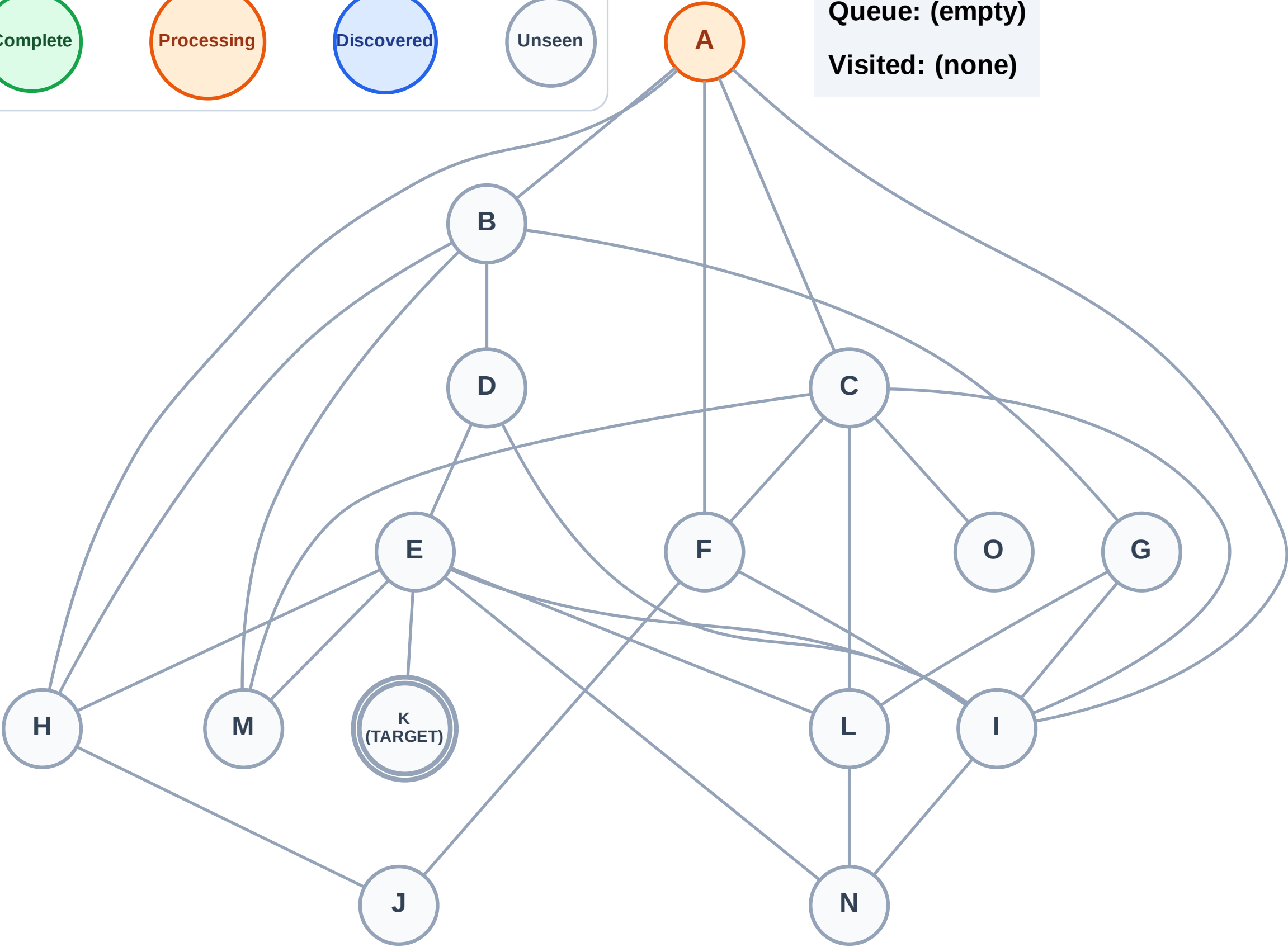
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, F, H, I from A

Legend

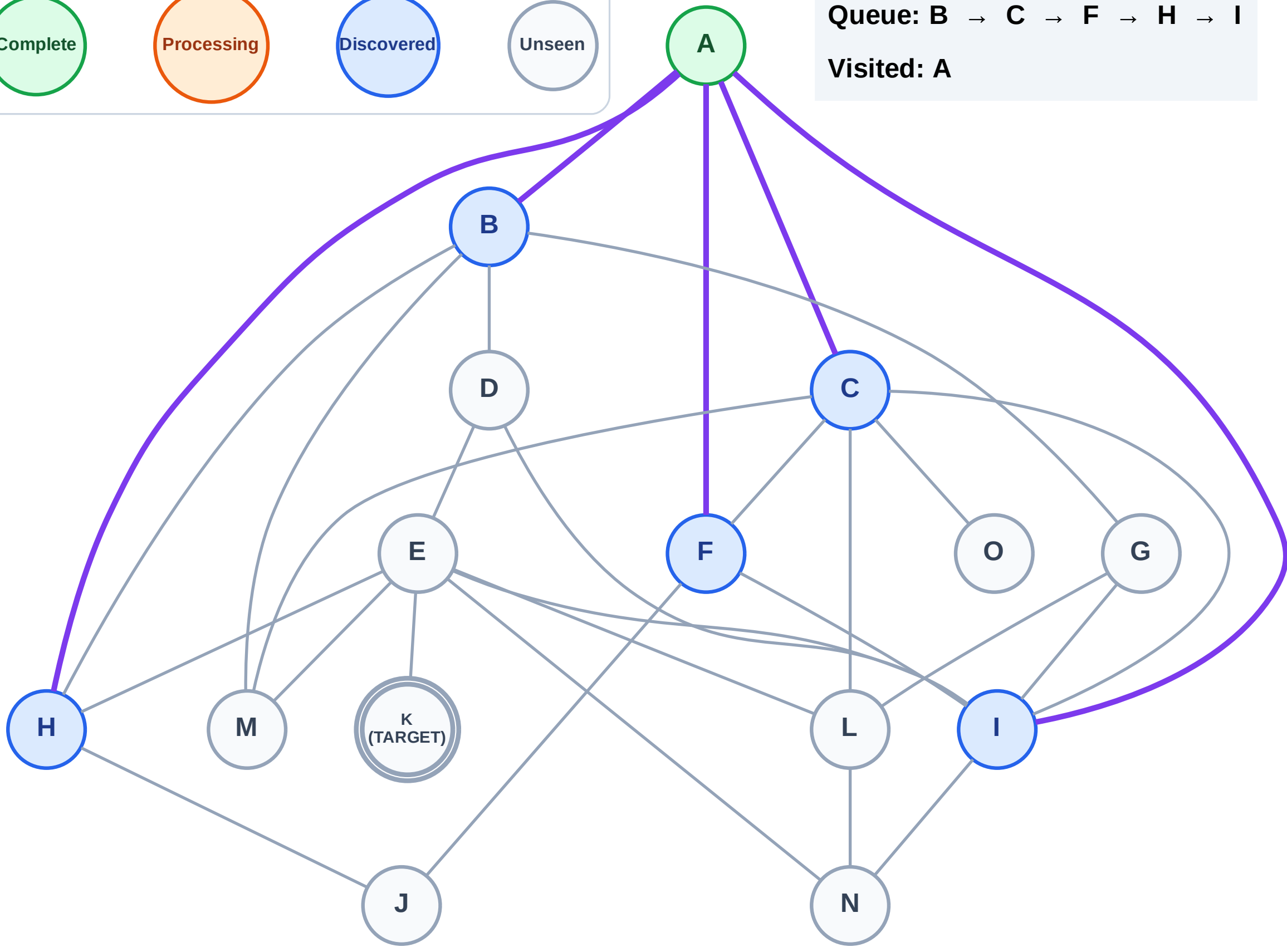
Complete

Processing

Discovered

Unseen

Queue: B → C → F → H → I
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

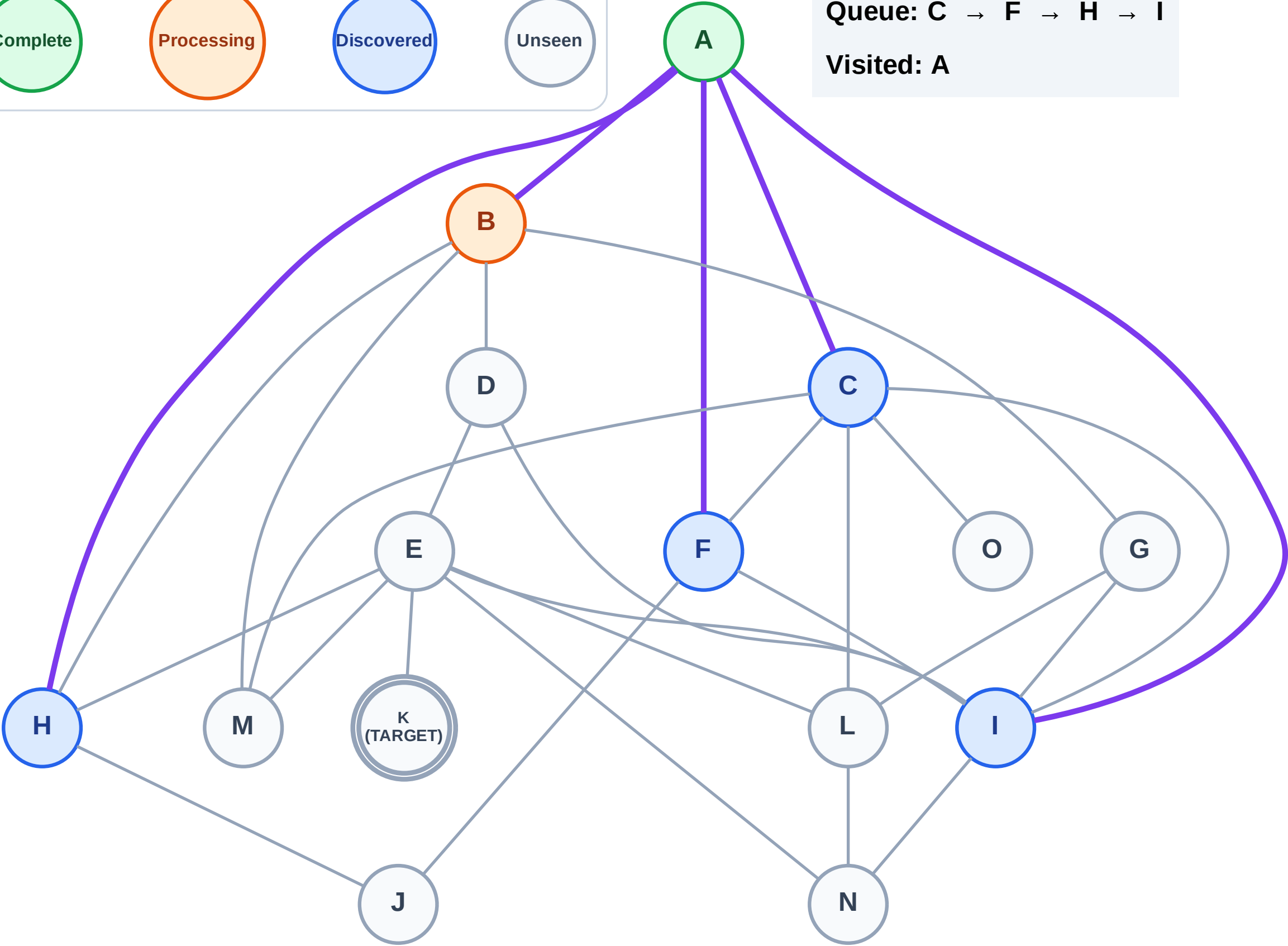
Processing

Discovered

Unseen

Queue: C → F → H → I

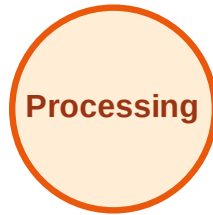
Visited: A



BFS Traversal

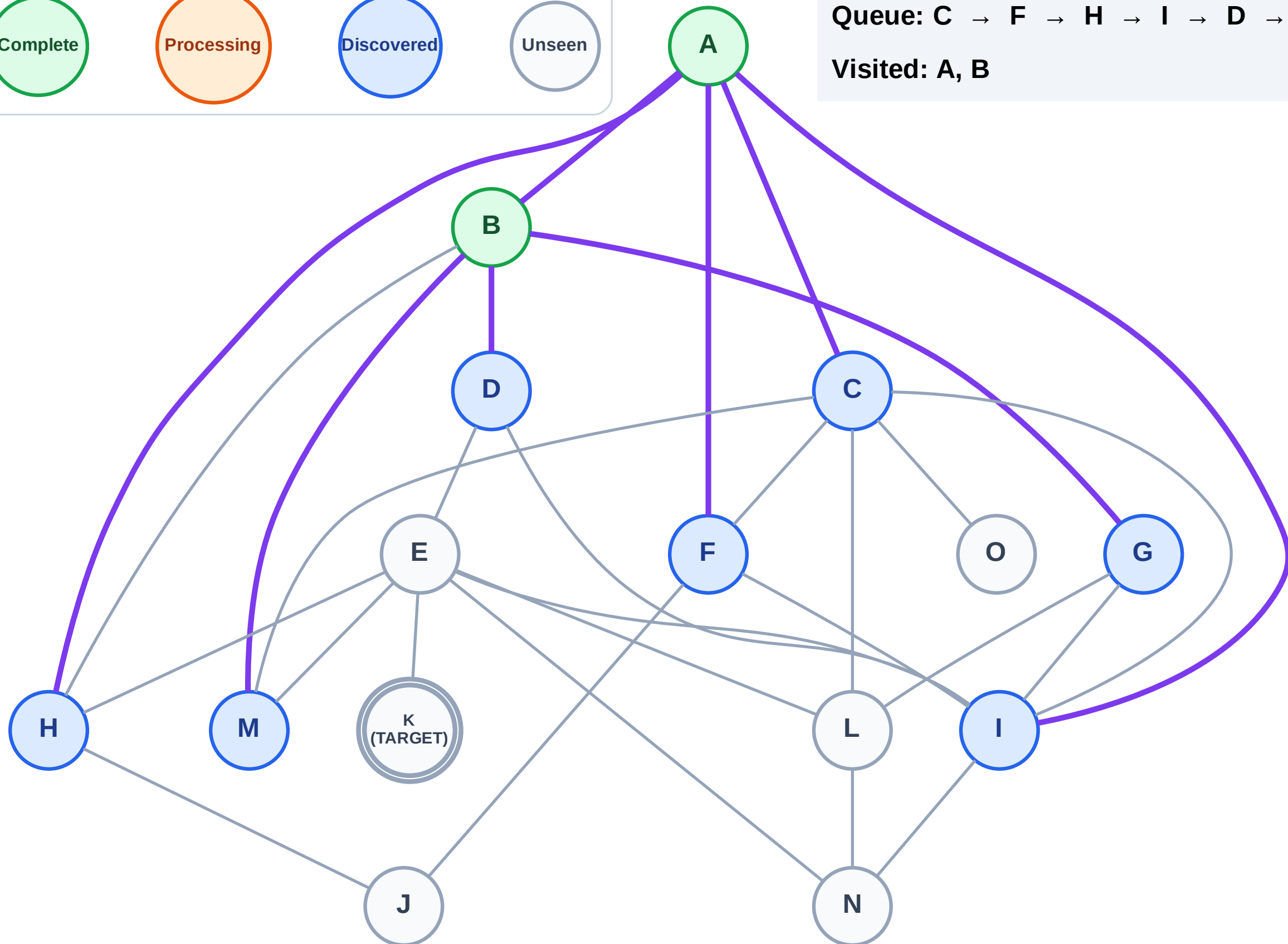
Step 5: Discover D, G, M from B

Legend



Queue: C → F → H → I → D → G → M

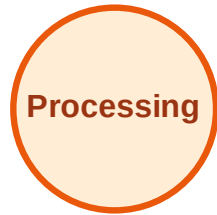
Visited: A, B



BFS Traversal

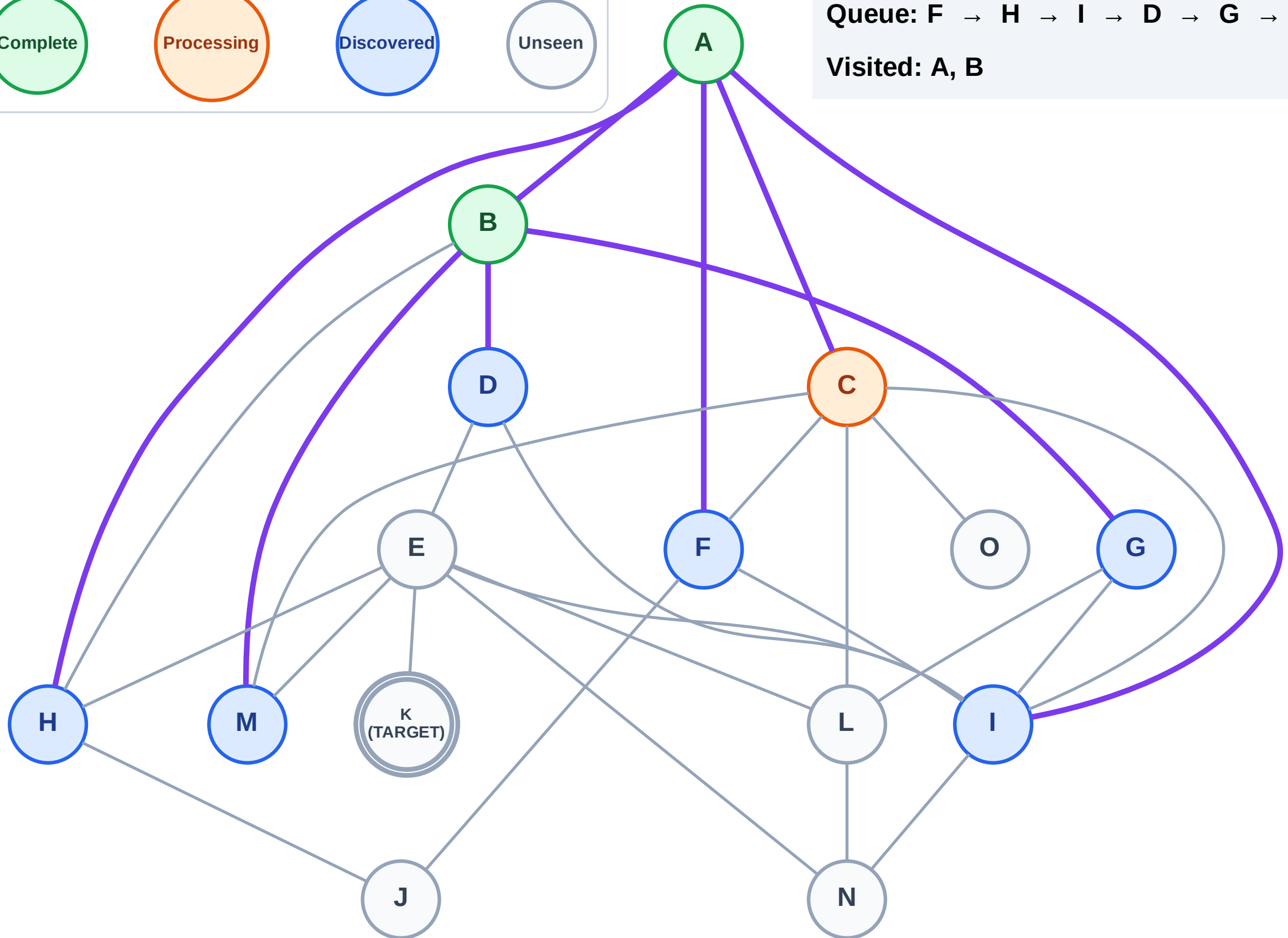
Step 6: Process node C

Legend



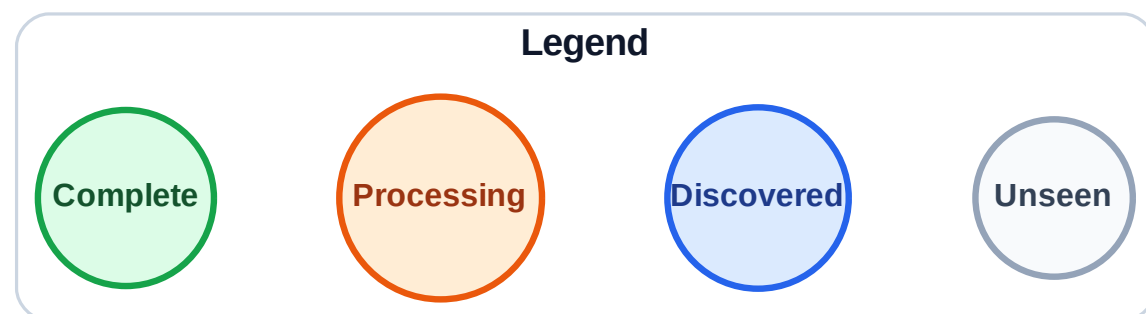
Queue: F → H → I → D → G → M

Visited: A, B



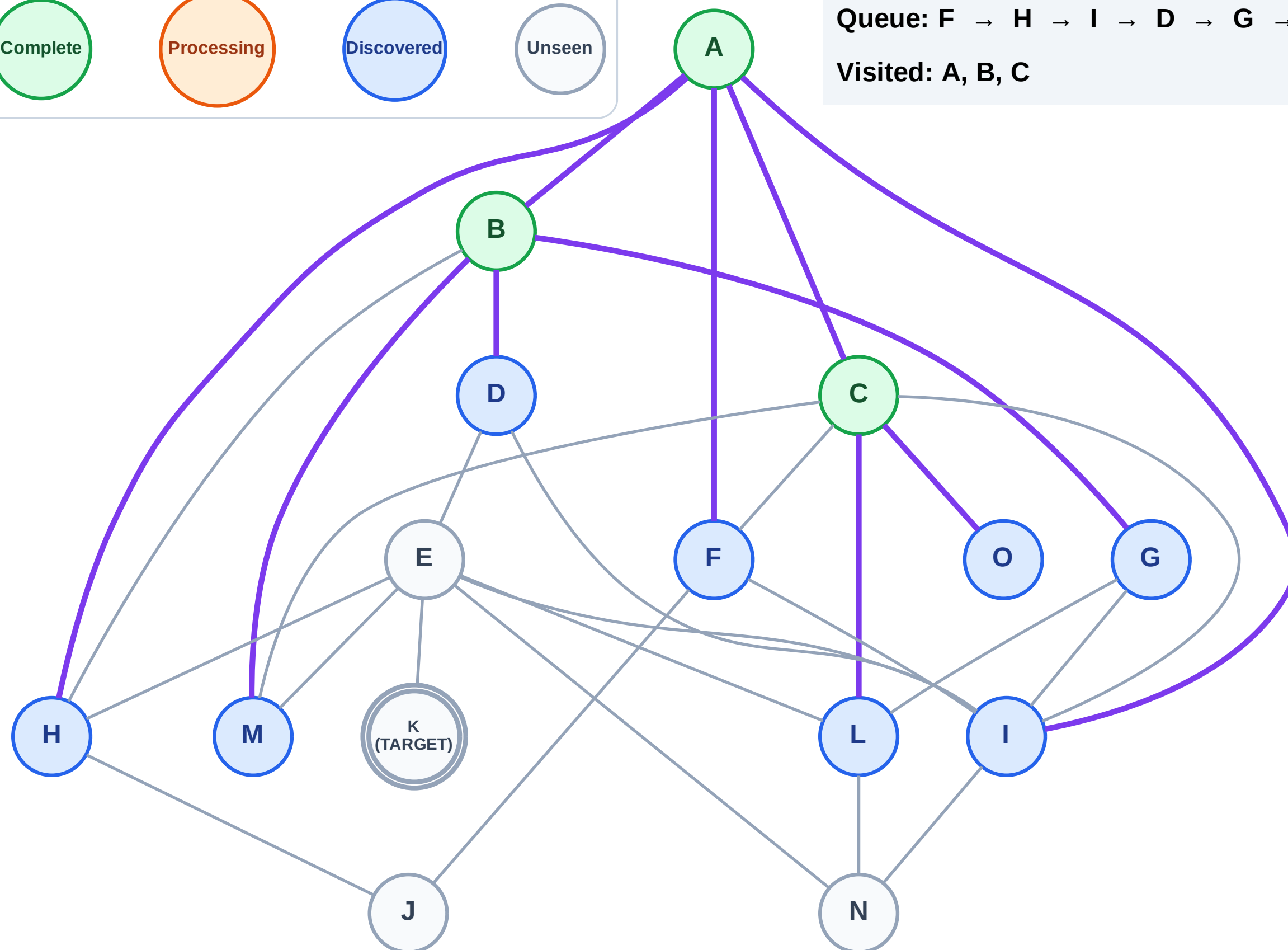
BFS Traversal

Step 7: Discover L, O from C



Queue: F → H → I → D → G → M → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

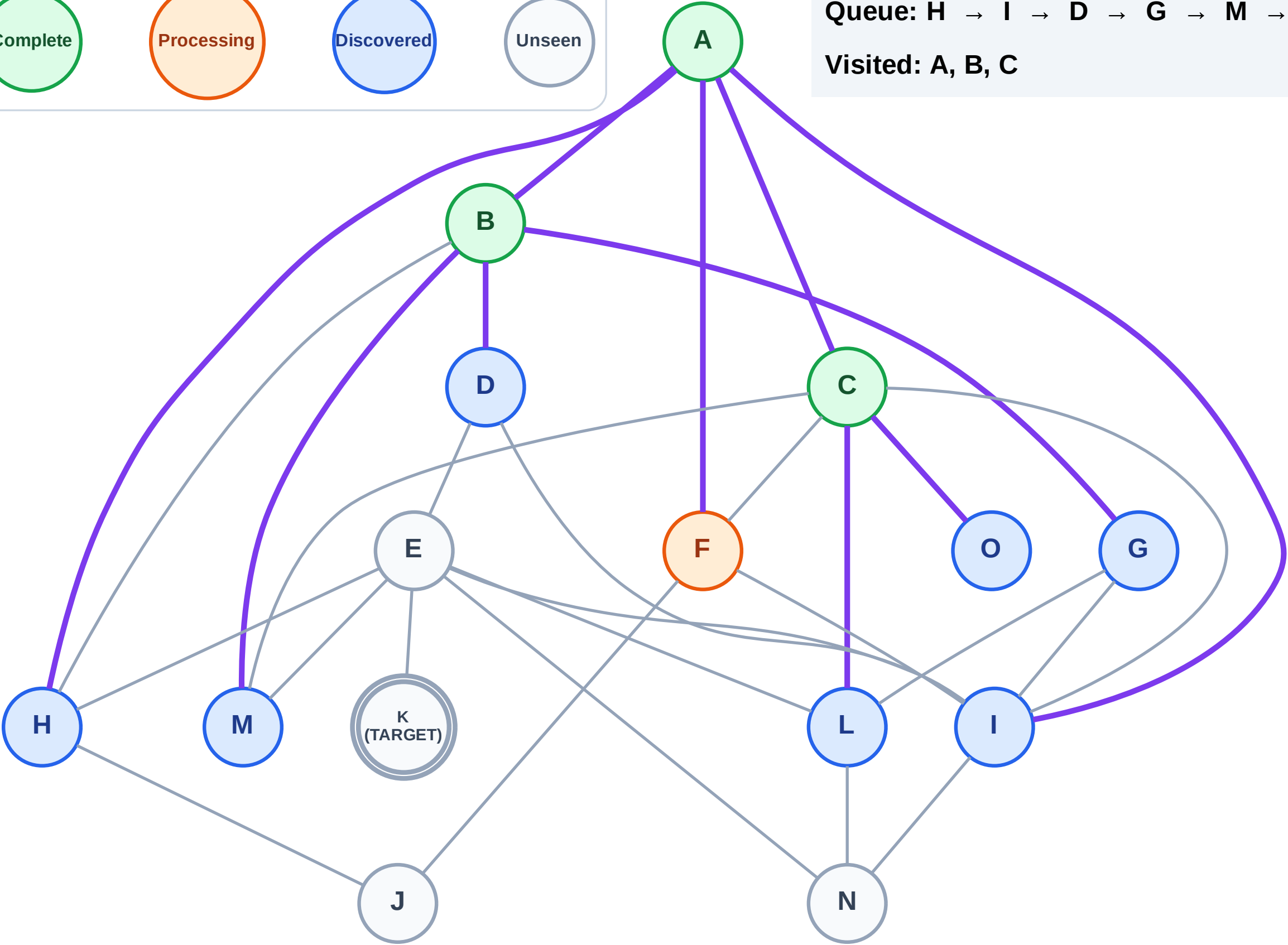
Processing

Discovered

Unseen

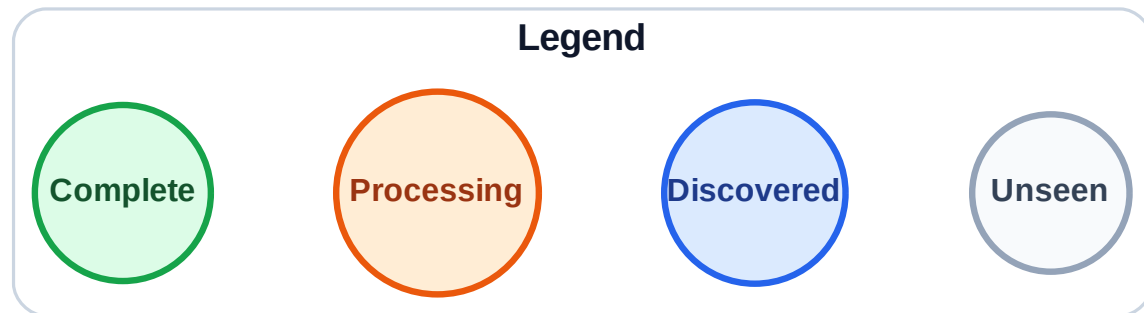
Queue: H → I → D → G → M → L → O

Visited: A, B, C



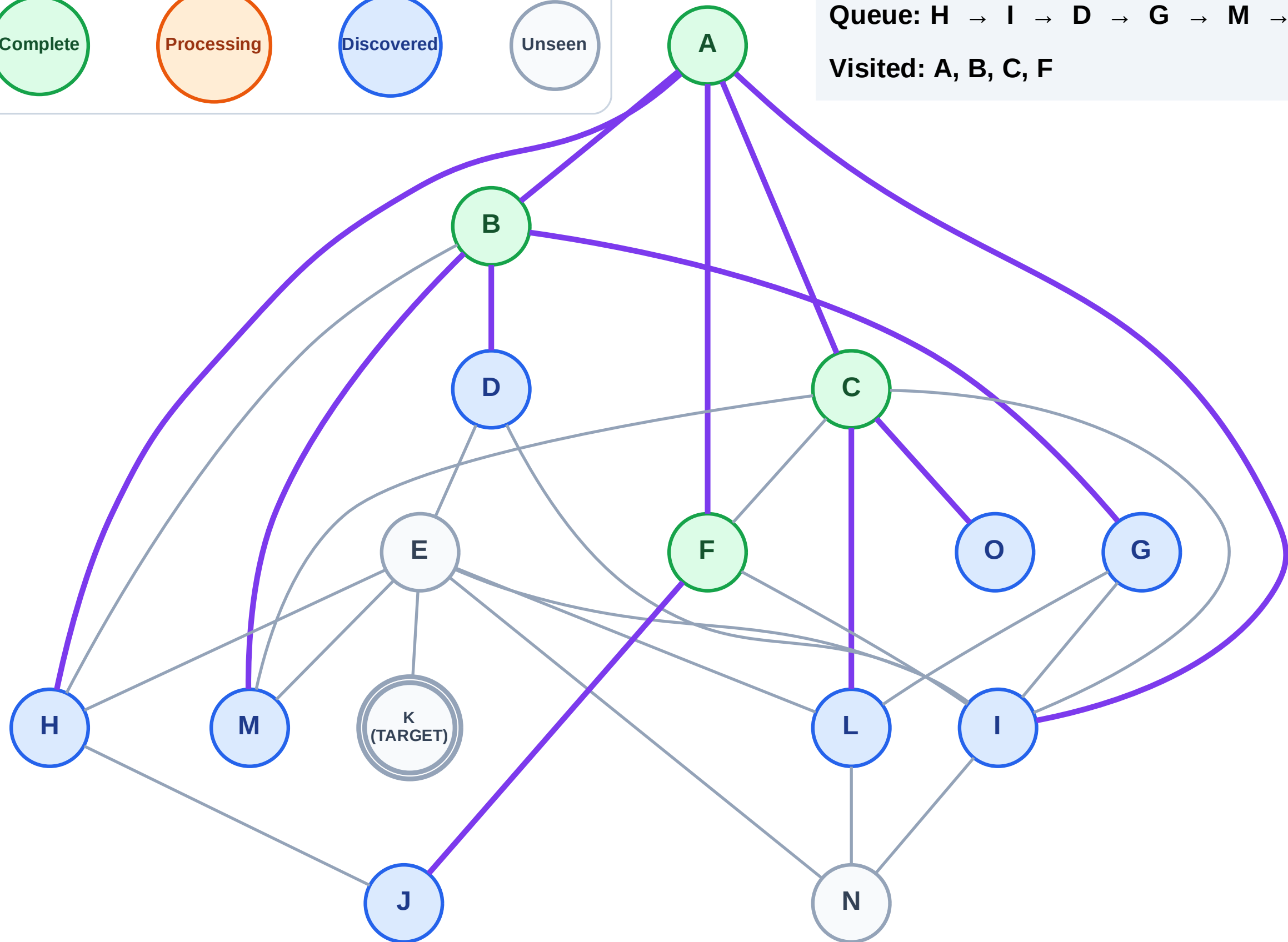
BFS Traversal

Step 9: Discover J from F



Queue: H → I → D → G → M → L → O → J

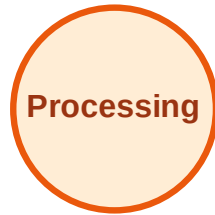
Visited: A, B, C, F



BFS Traversal

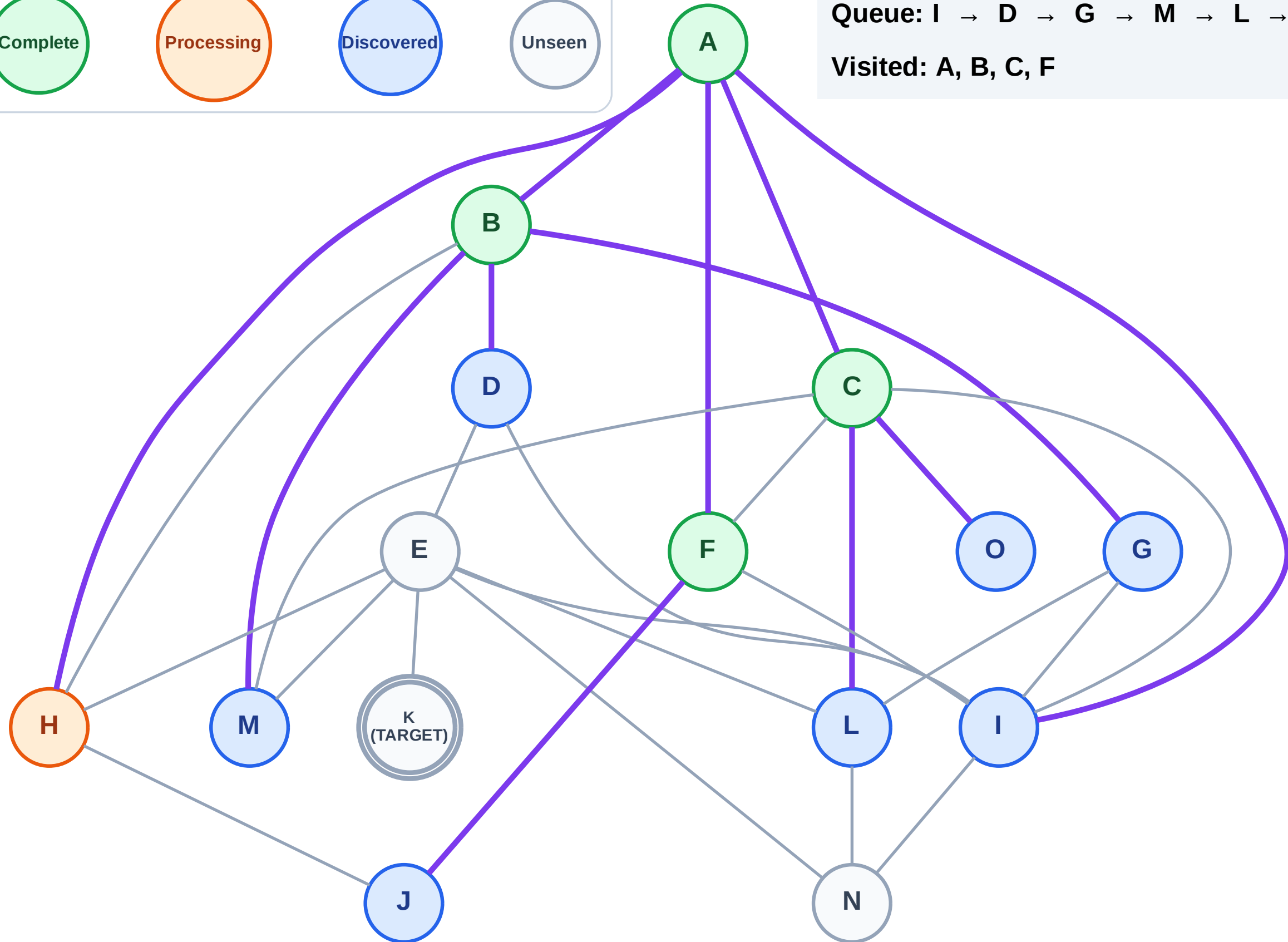
Step 10: Process node H

Legend



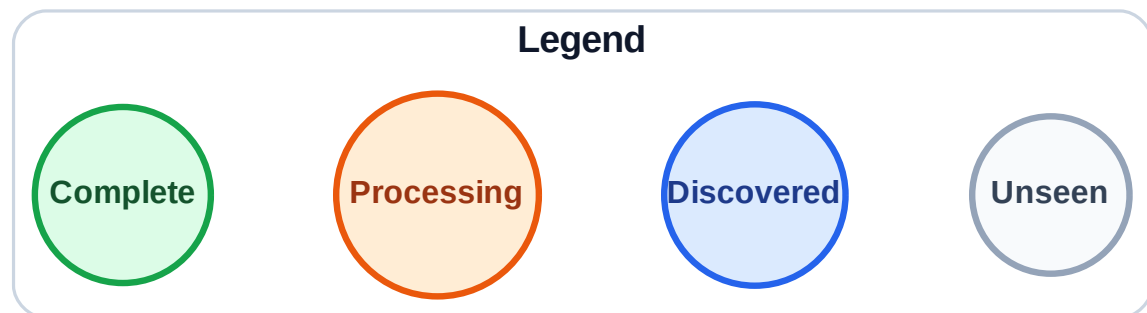
Queue: I → D → G → M → L → O → J

Visited: A, B, C, F



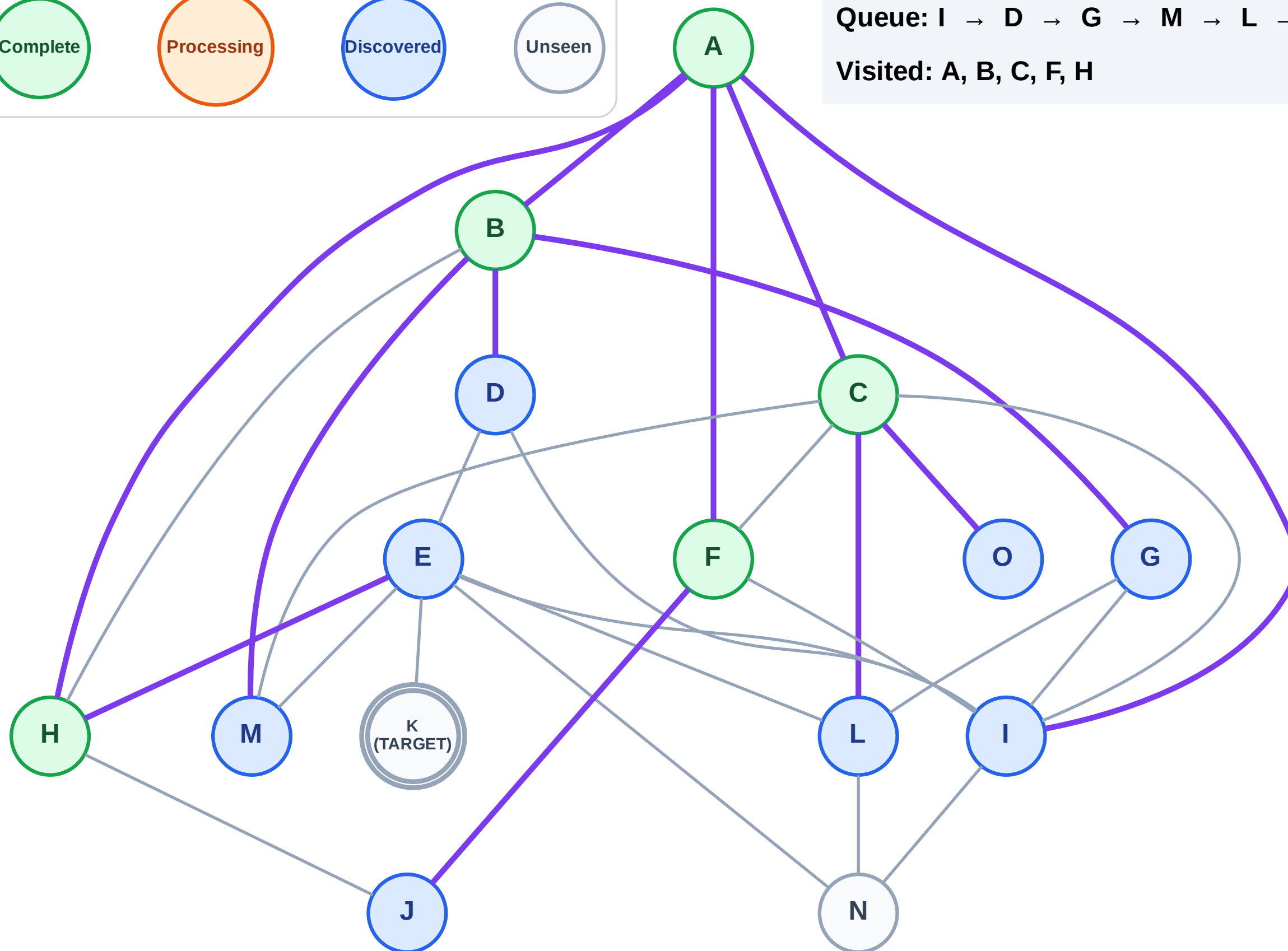
BFS Traversal

Step 11: Discover E from H



Queue: I → D → G → M → L → O → J → E

Visited: A, B, C, F, H



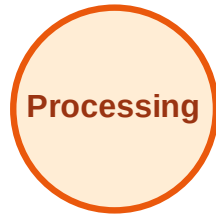
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



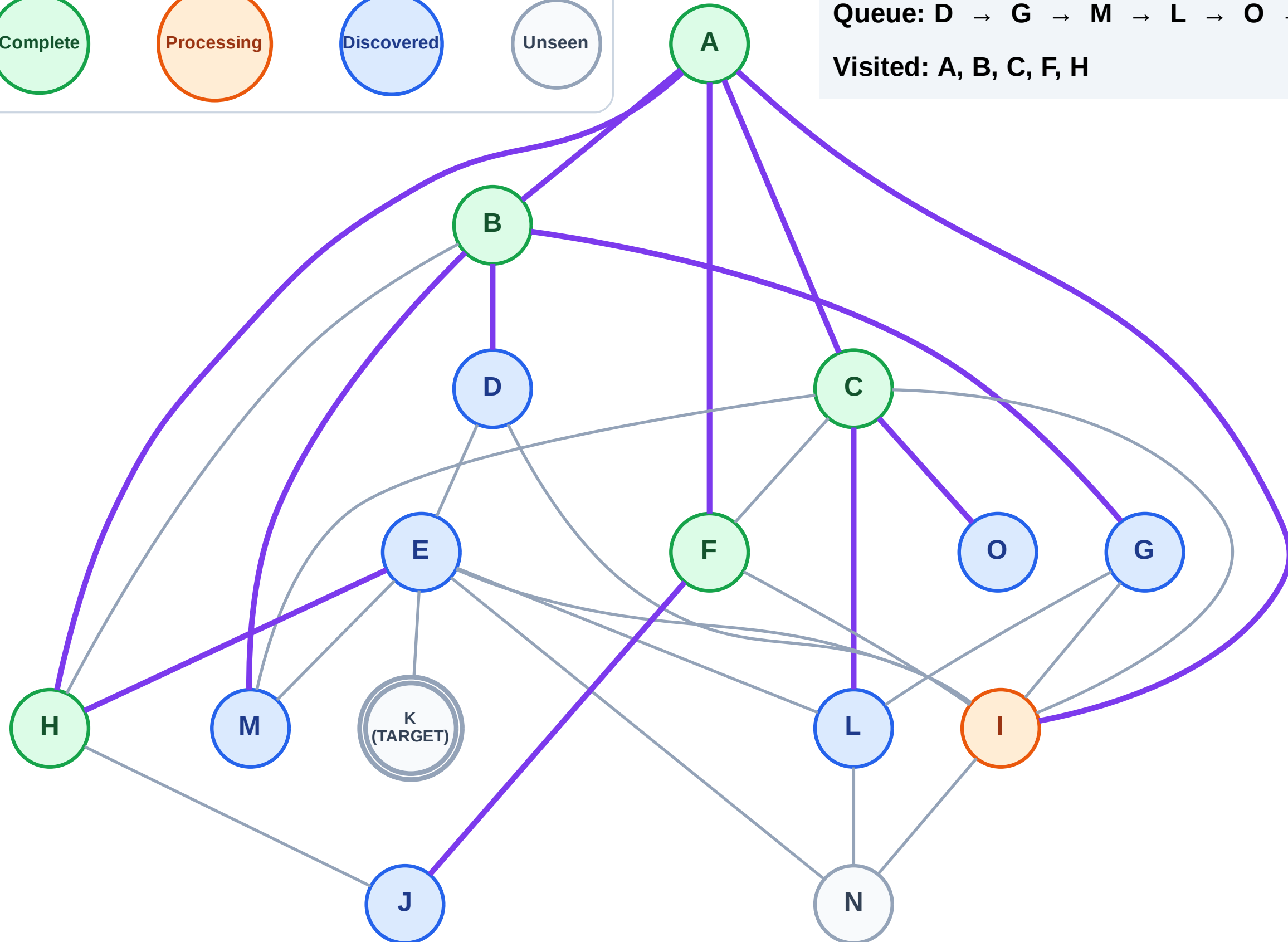
Discovered



Unseen

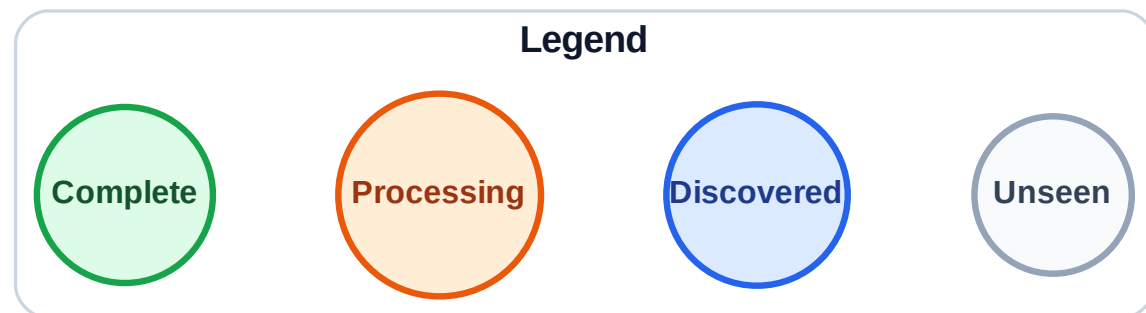
Queue: D → G → M → L → O → J → E

Visited: A, B, C, F, H



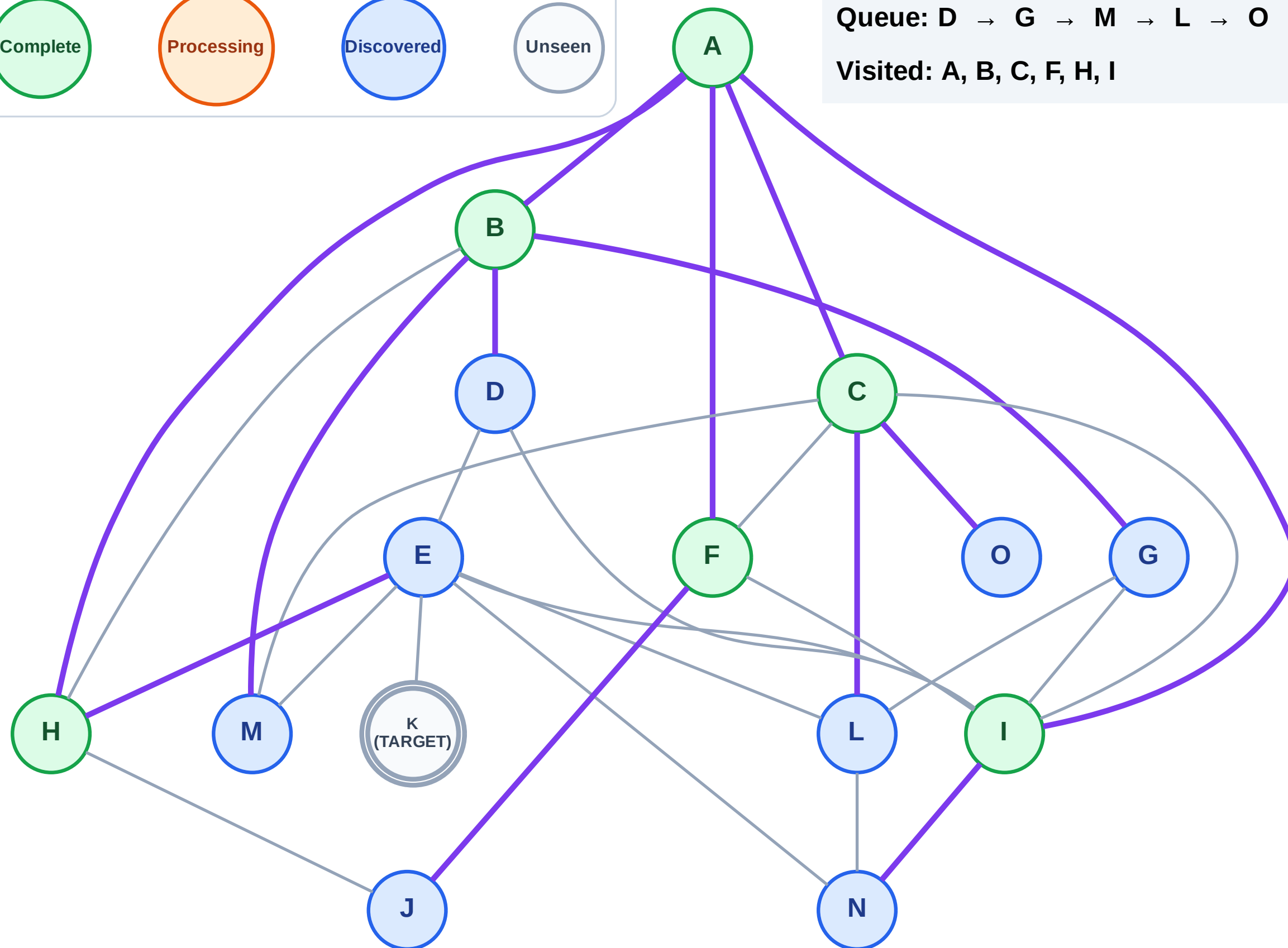
BFS Traversal

Step 13: Discover N from I



Queue: D → G → M → L → O → J → E → N

Visited: A, B, C, F, H, I



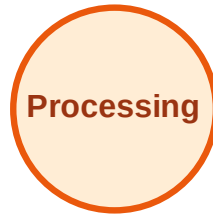
BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



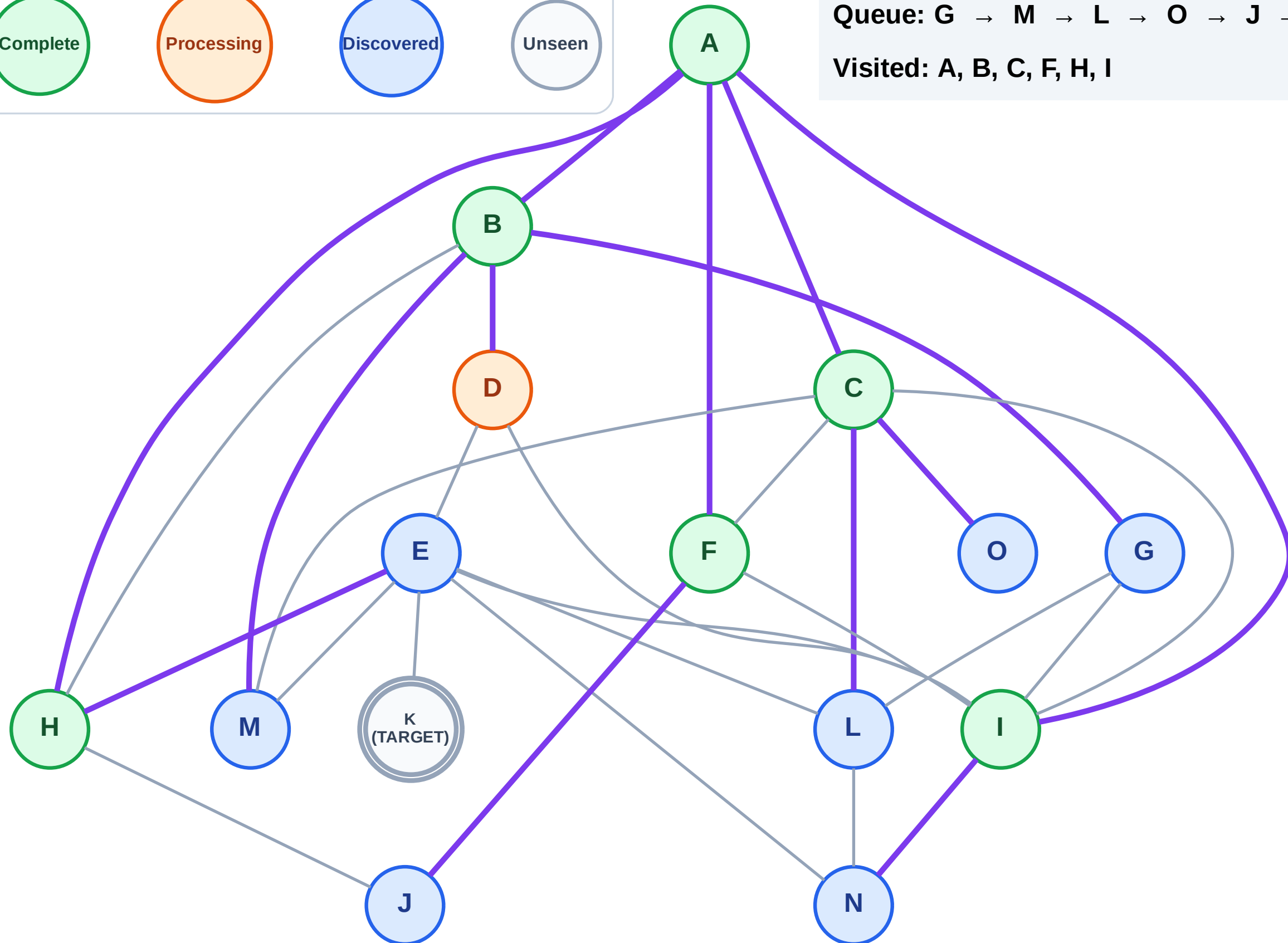
Discovered



Unseen

Queue: G → M → L → O → J → E → N

Visited: A, B, C, F, H, I



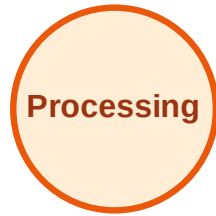
BFS Traversal

Step 15: No new neighbors from D

Legend



Complete



Processing



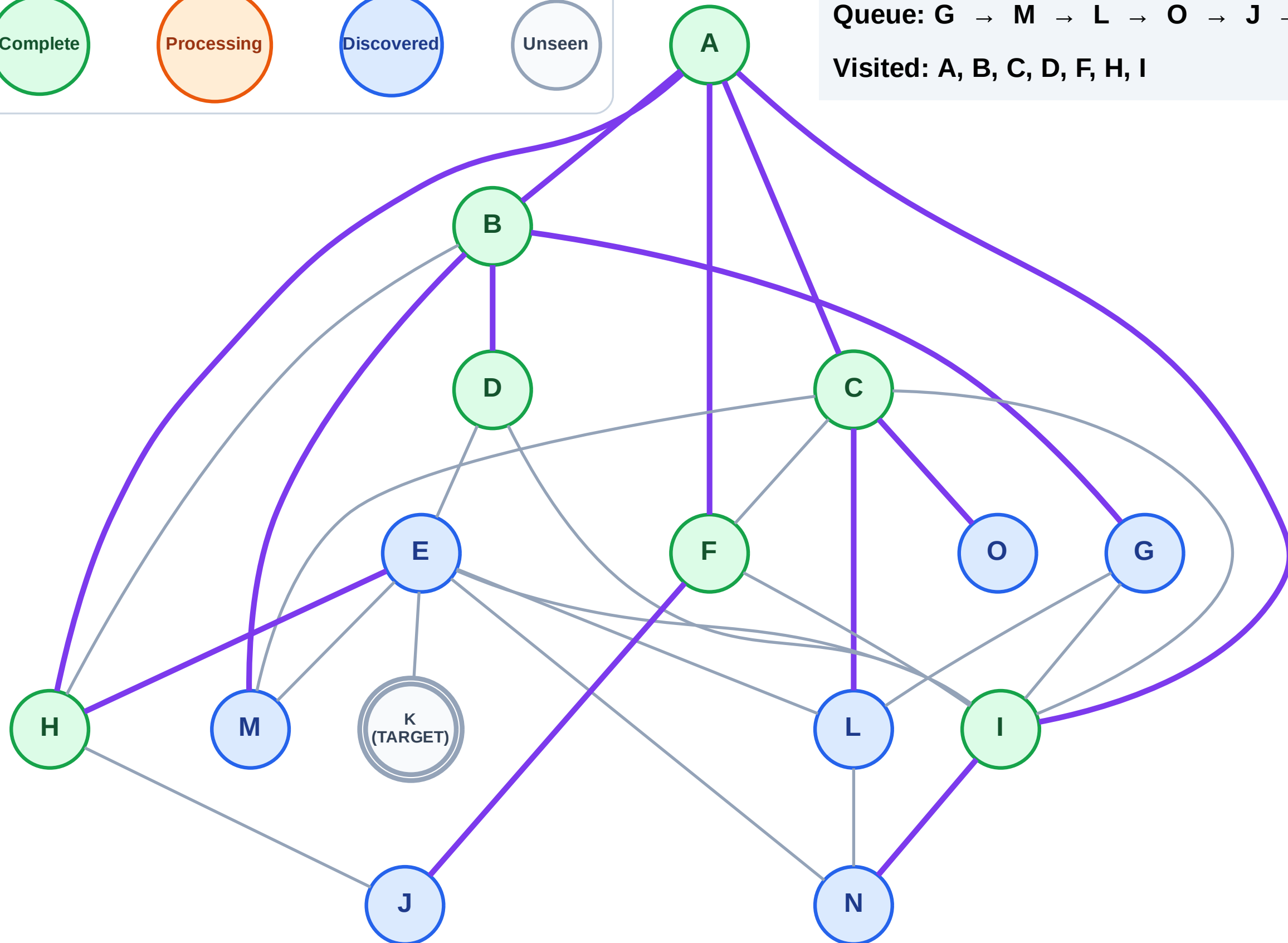
Discovered



Unseen

Queue: G → M → L → O → J → E → N

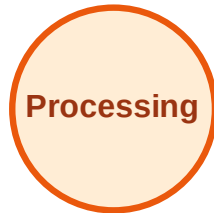
Visited: A, B, C, D, F, H, I



BFS Traversal

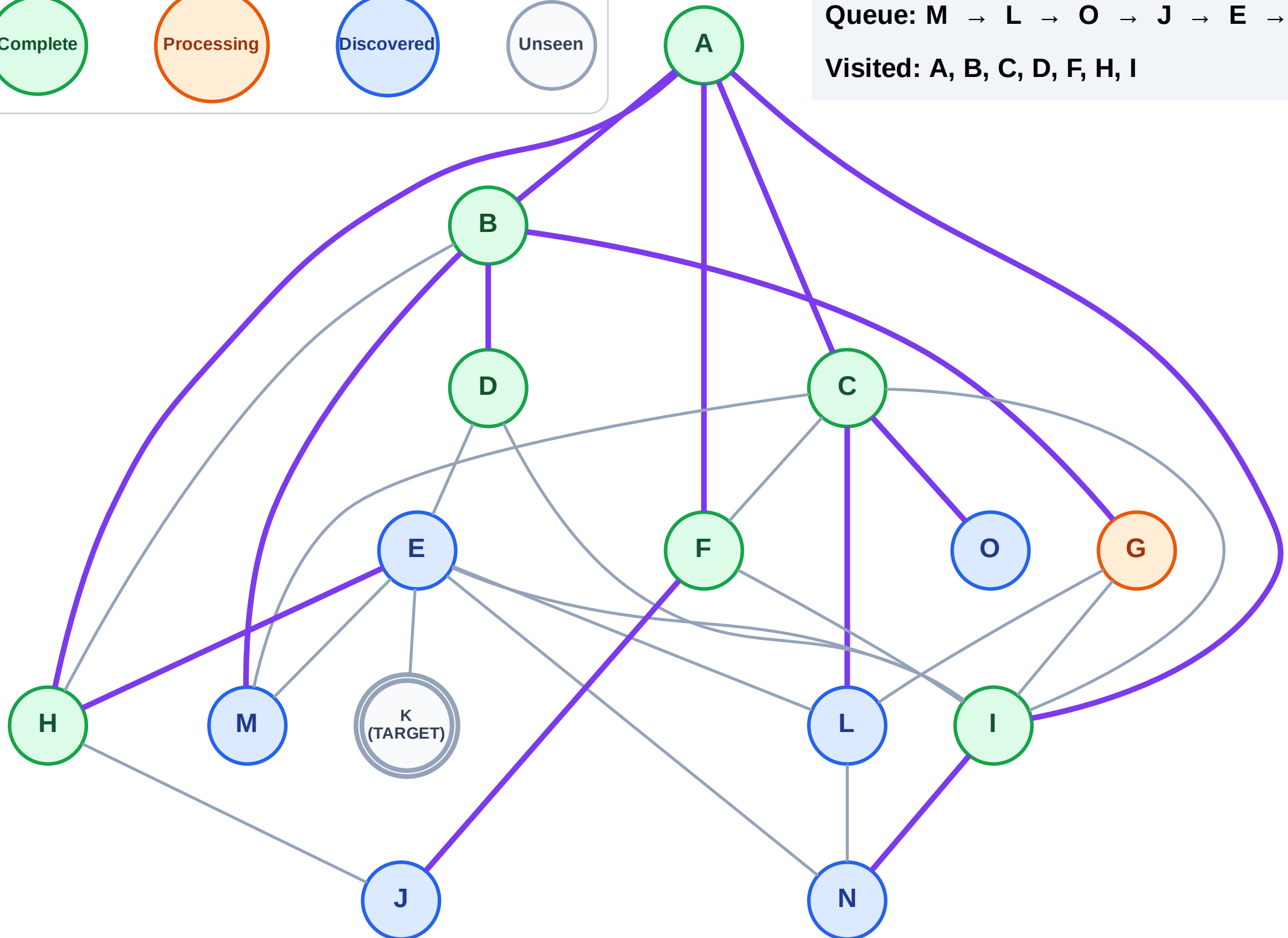
Step 16: Process node G

Legend



Queue: M → L → O → J → E → N

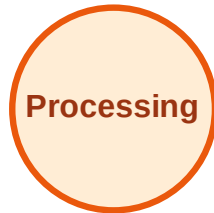
Visited: A, B, C, D, F, H, I



BFS Traversal

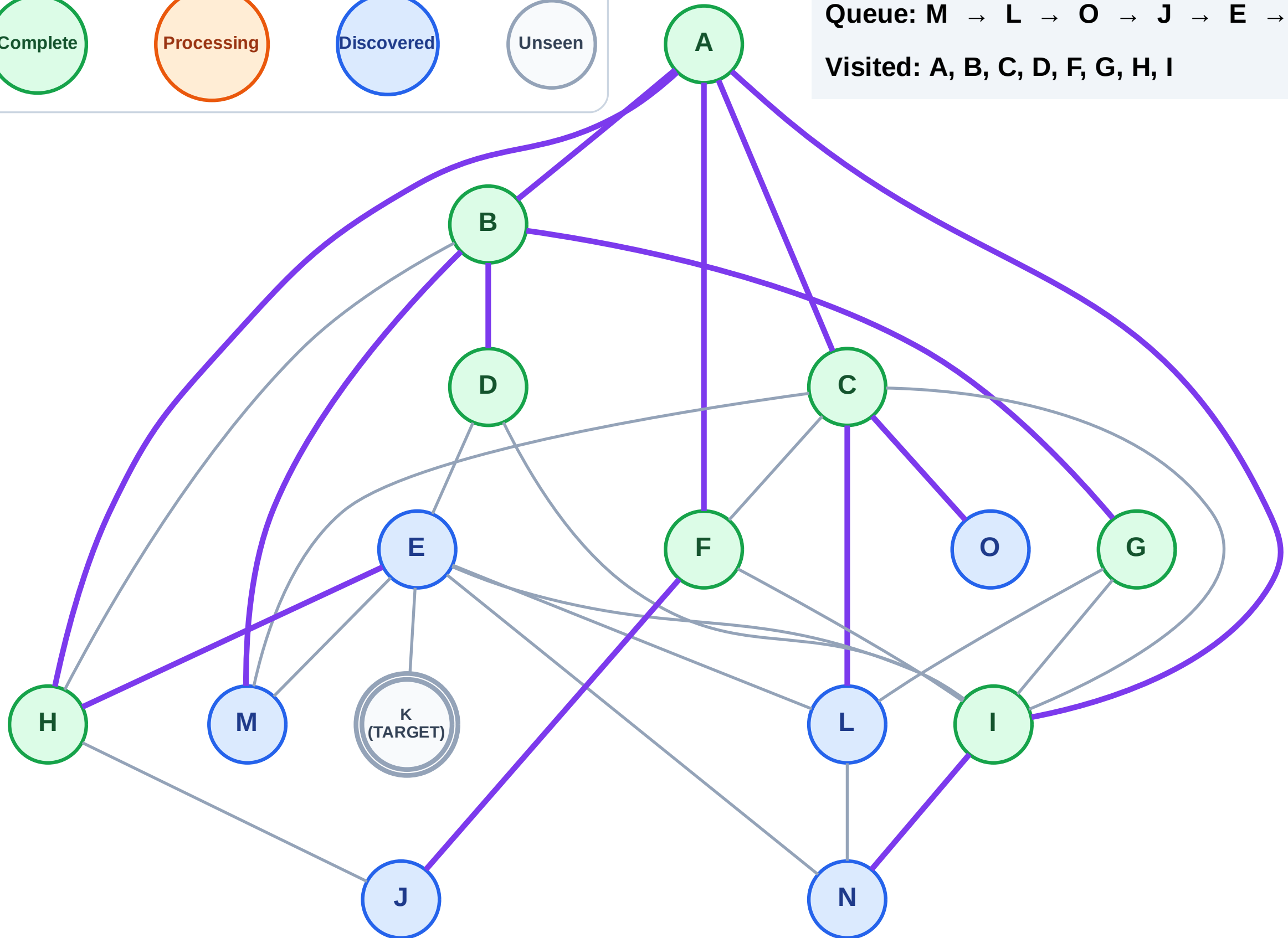
Step 17: No new neighbors from G

Legend



Queue: M → L → O → J → E → N

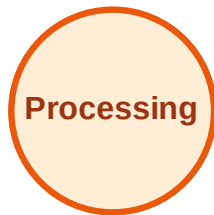
Visited: A, B, C, D, F, G, H, I



BFS Traversal

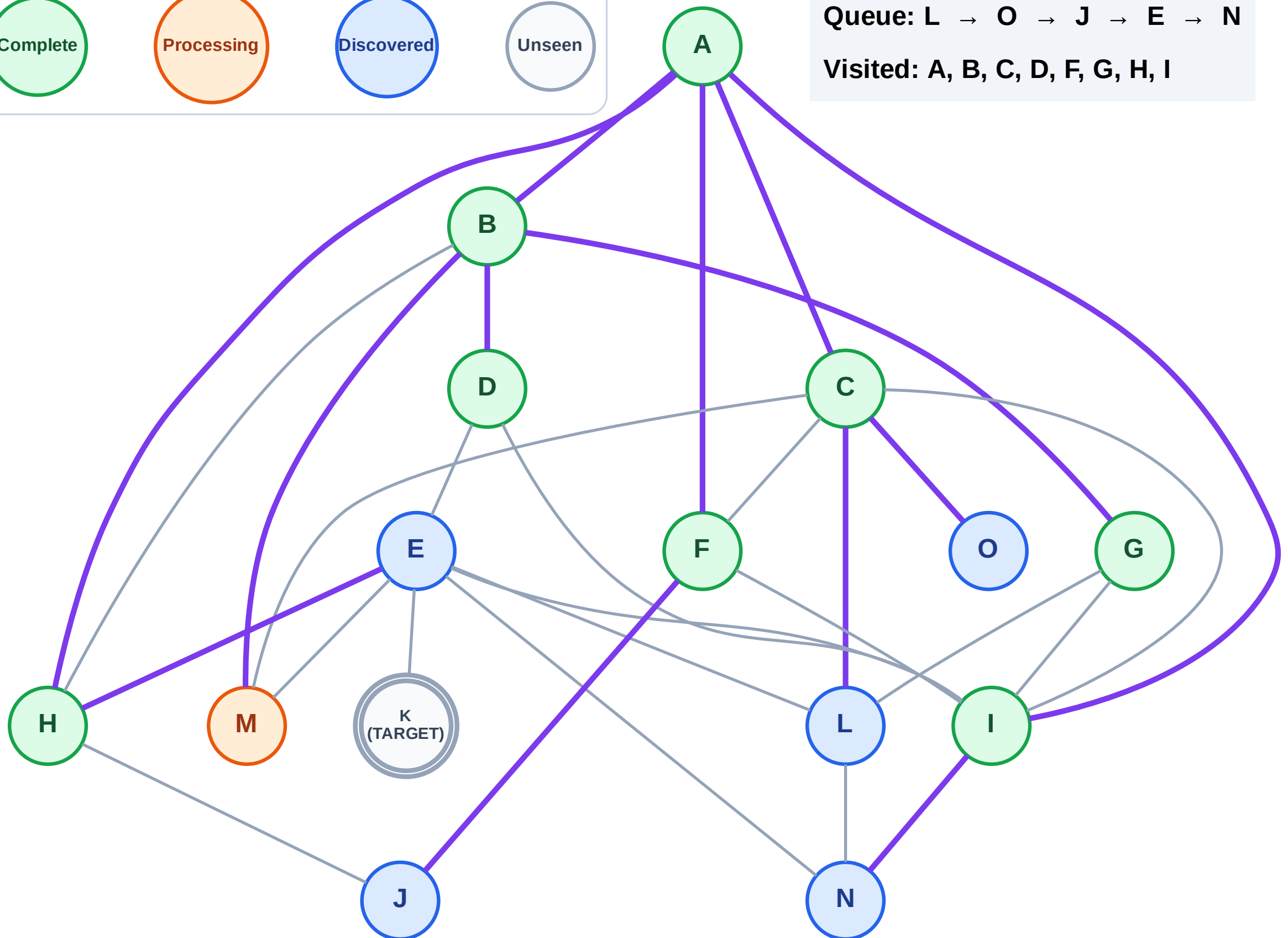
Step 18: Process node M

Legend



Queue: L → O → J → E → N

Visited: A, B, C, D, F, G, H, I



BFS Traversal

Step 19: No new neighbors from M

Legend

Complete

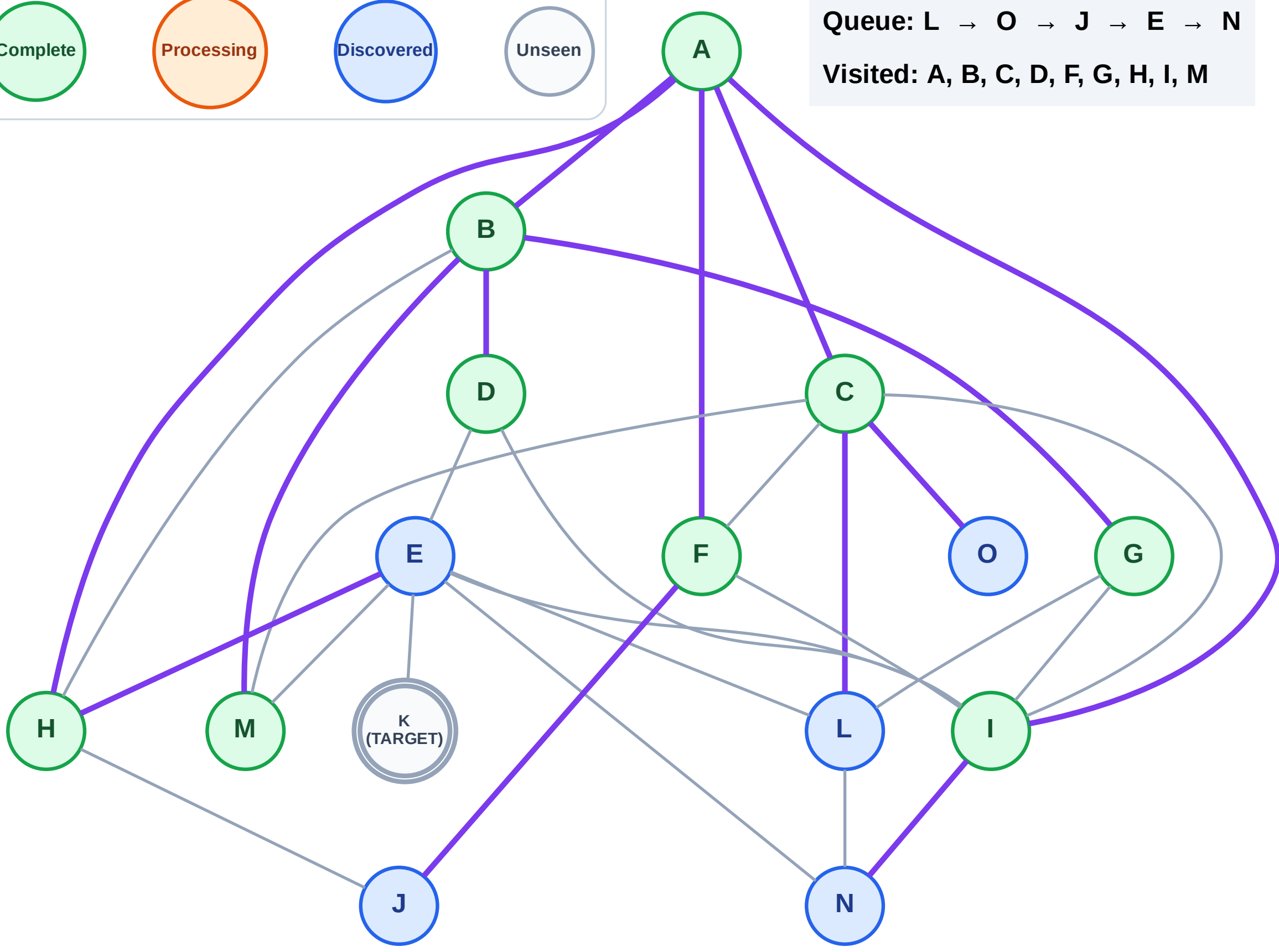
Processing

Discovered

Unseen

Queue: L → O → J → E → N

Visited: A, B, C, D, F, G, H, I, M



BFS Traversal

Step 20: Process node L

Legend

Complete

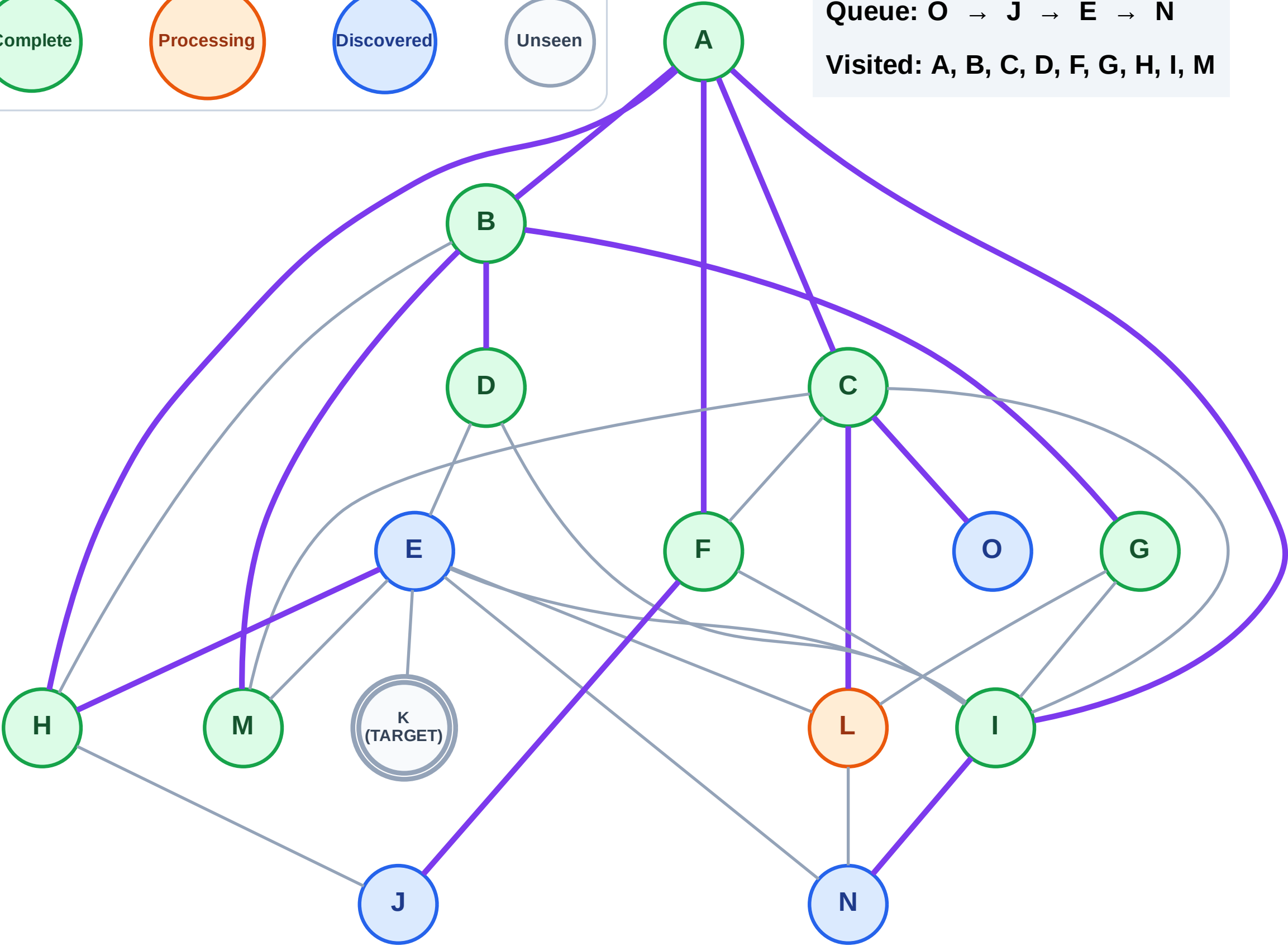
Processing

Discovered

Unseen

Queue: O → J → E → N

Visited: A, B, C, D, F, G, H, I, M



BFS Traversal

Step 21: No new neighbors from L

Legend

Complete

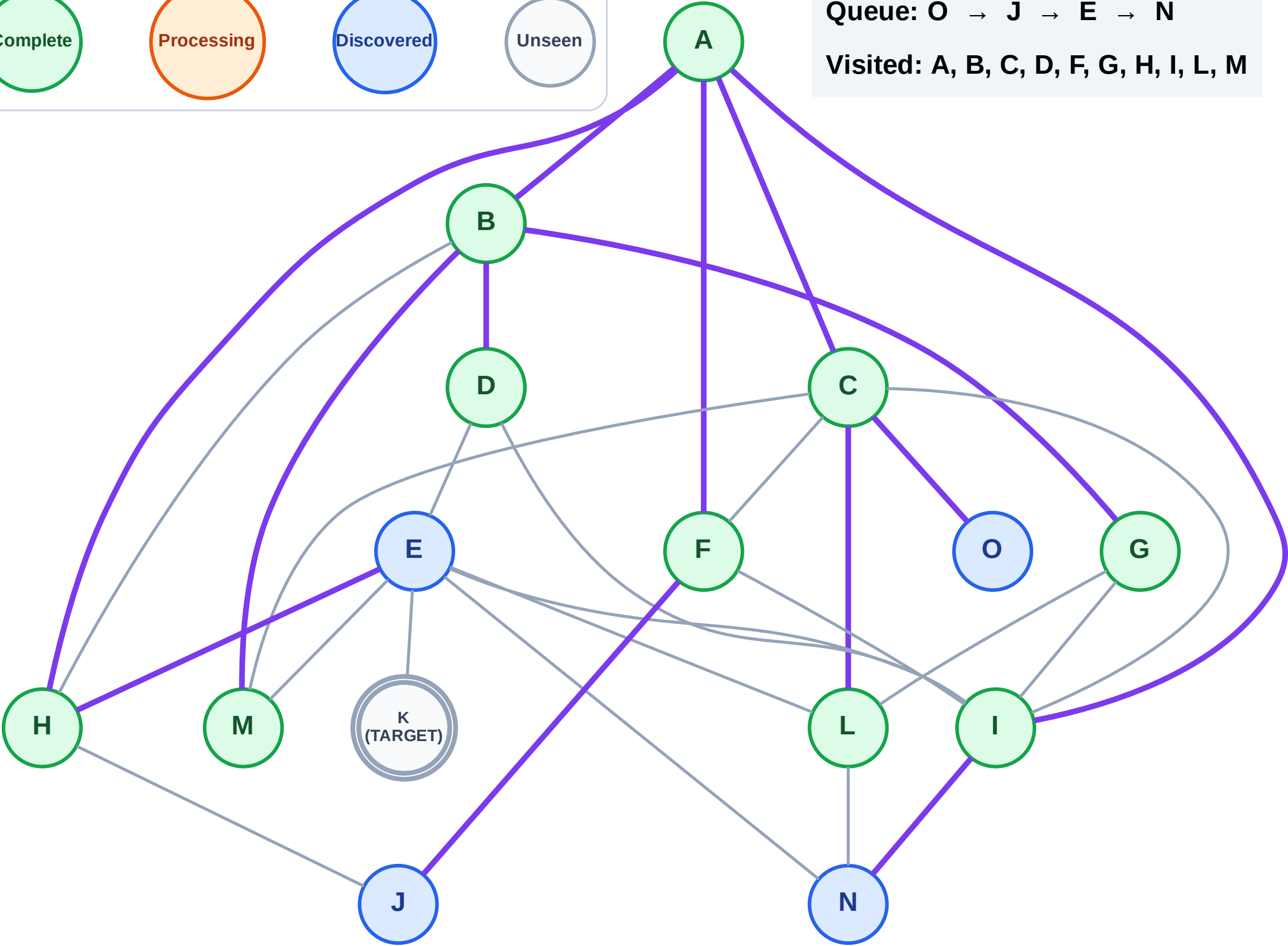
Processing

Discovered

Unseen

Queue: O → J → E → N

Visited: A, B, C, D, F, G, H, I, L, M



BFS Traversal

Step 22: Process node O

Legend

Complete

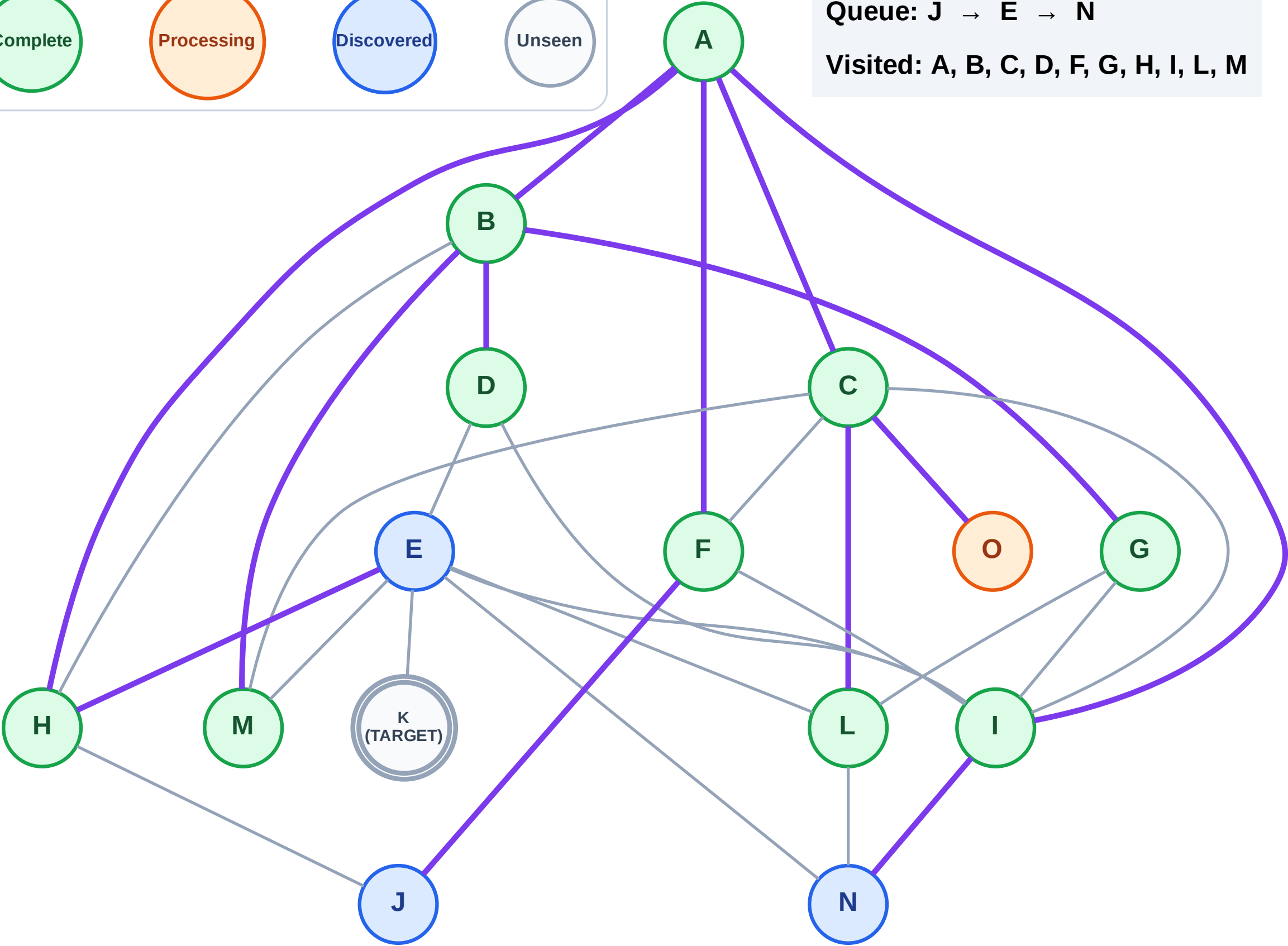
Processing

Discovered

Unseen

Queue: J → E → N

Visited: A, B, C, D, F, G, H, I, L, M



BFS Traversal

Step 23: No new neighbors from O

Legend

Complete

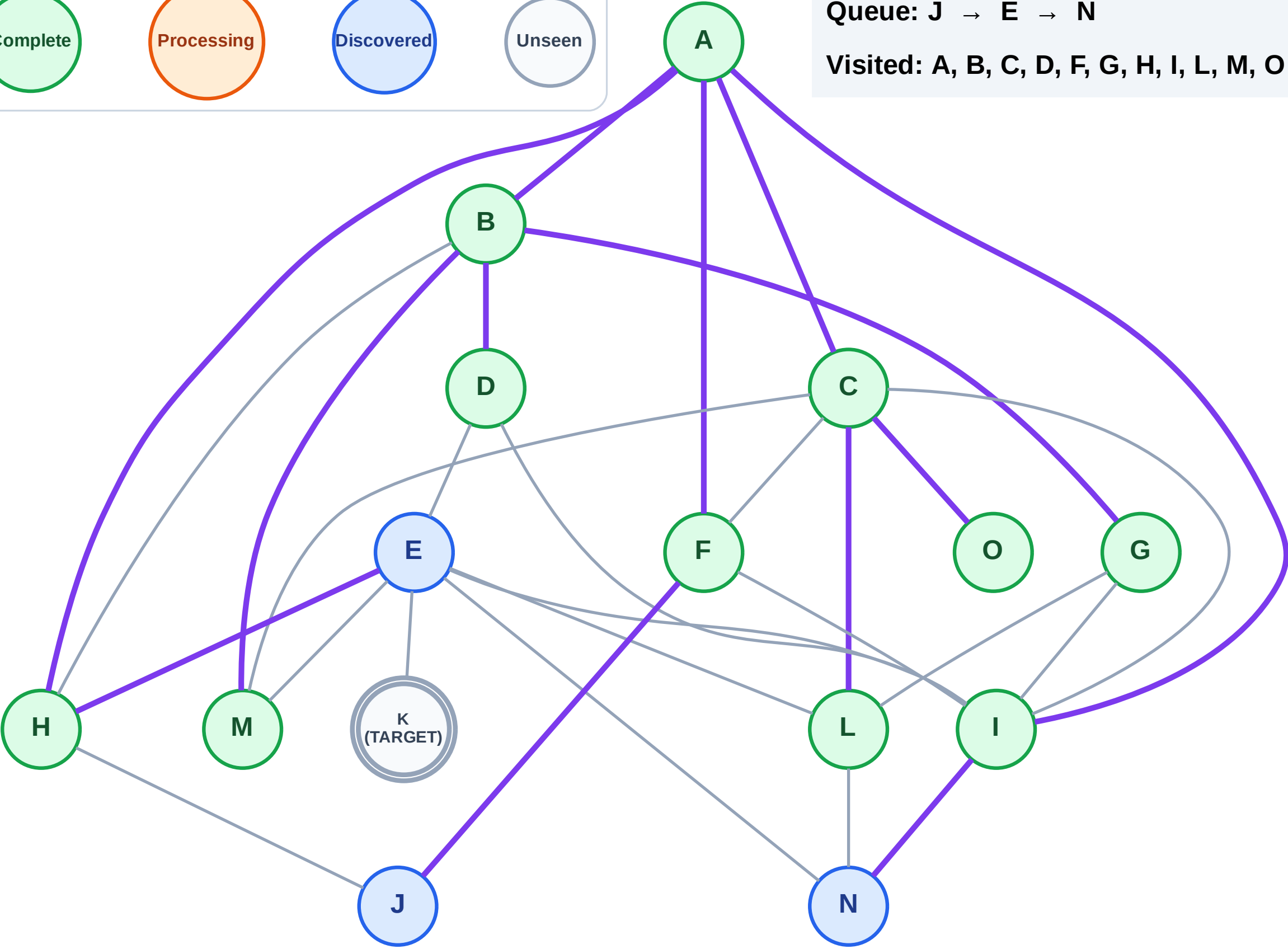
Processing

Discovered

Unseen

Queue: J → E → N

Visited: A, B, C, D, F, G, H, I, L, M, O



BFS Traversal

Step 24: Process node J

Legend

Complete

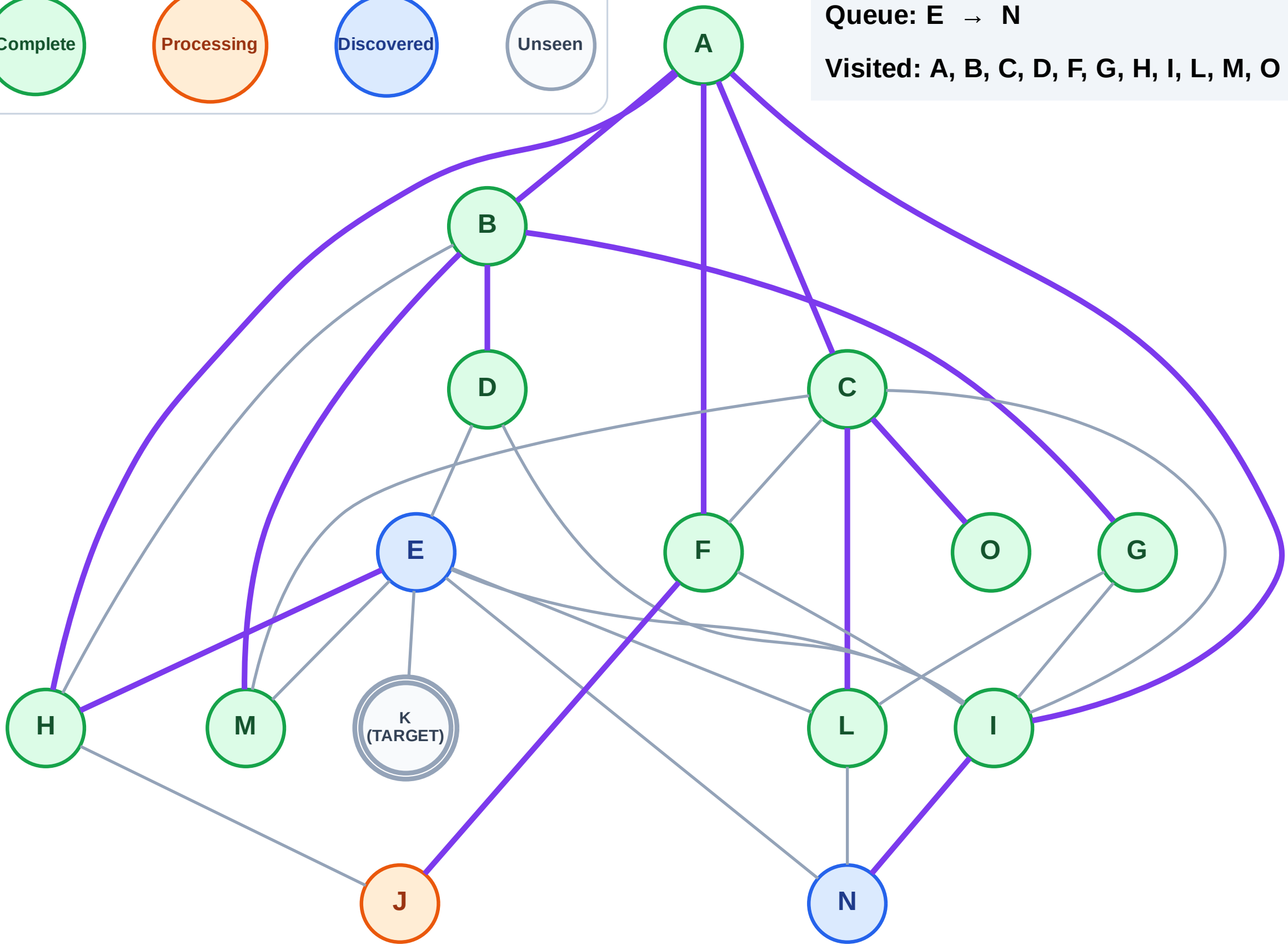
Processing

Discovered

Unseen

Queue: E → N

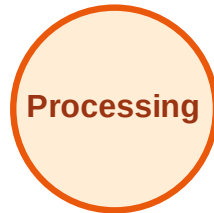
Visited: A, B, C, D, F, G, H, I, L, M, O



BFS Traversal

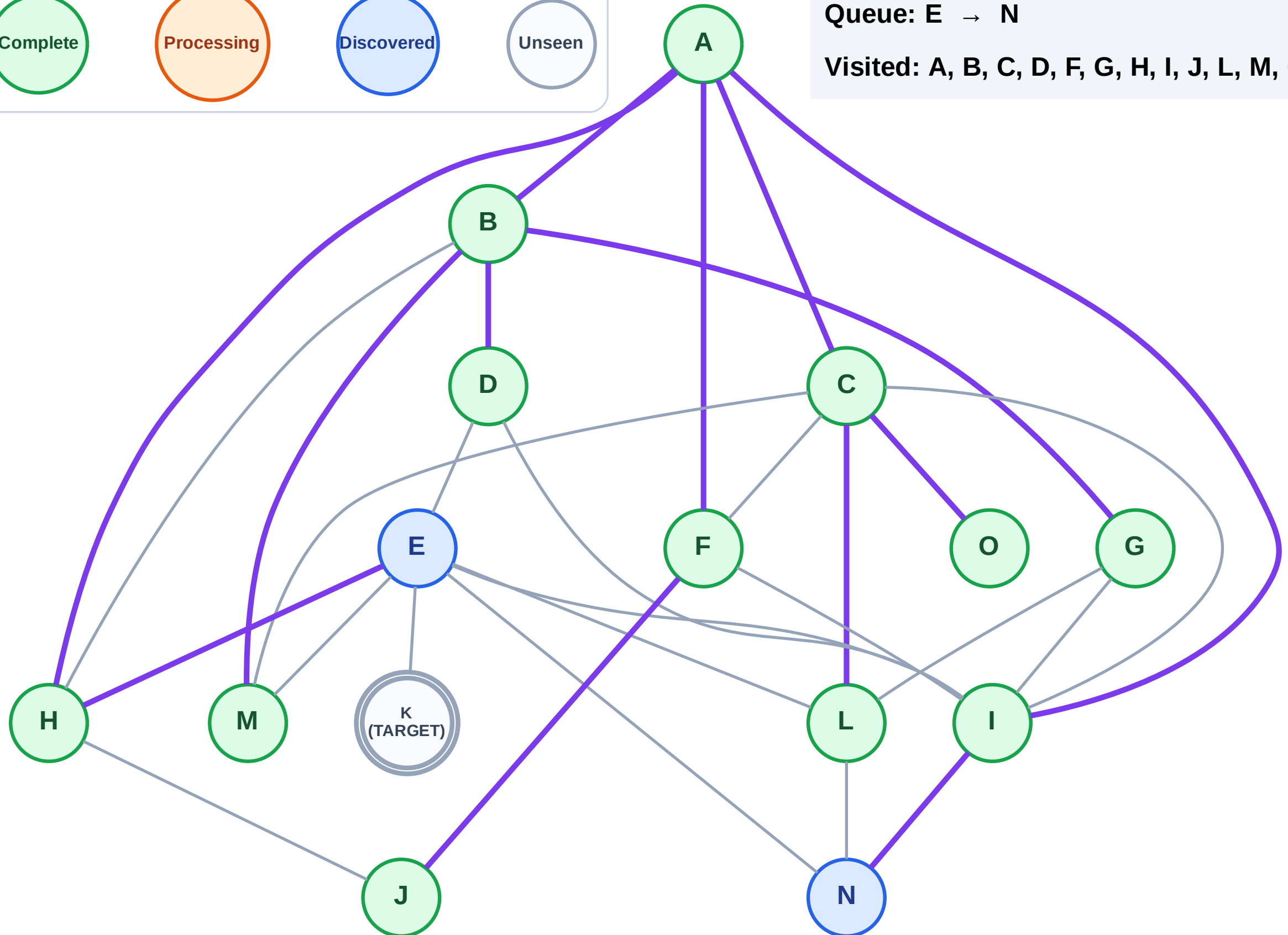
Step 25: No new neighbors from J

Legend



Queue: E → N

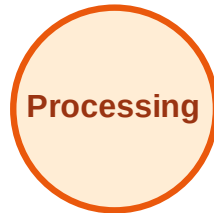
Visited: A, B, C, D, F, G, H, I, J, L, M, O



BFS Traversal

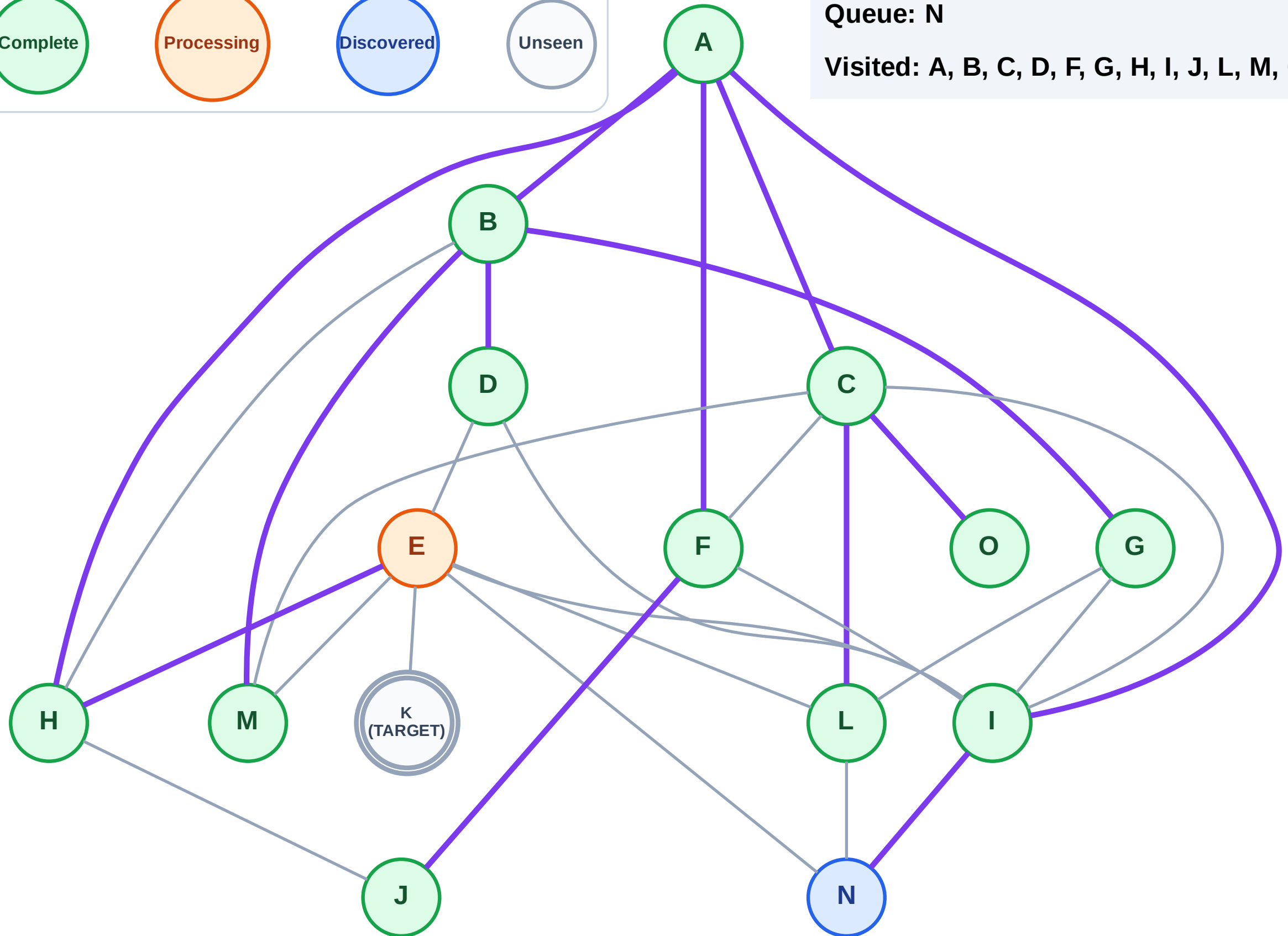
Step 26: Process node E

Legend



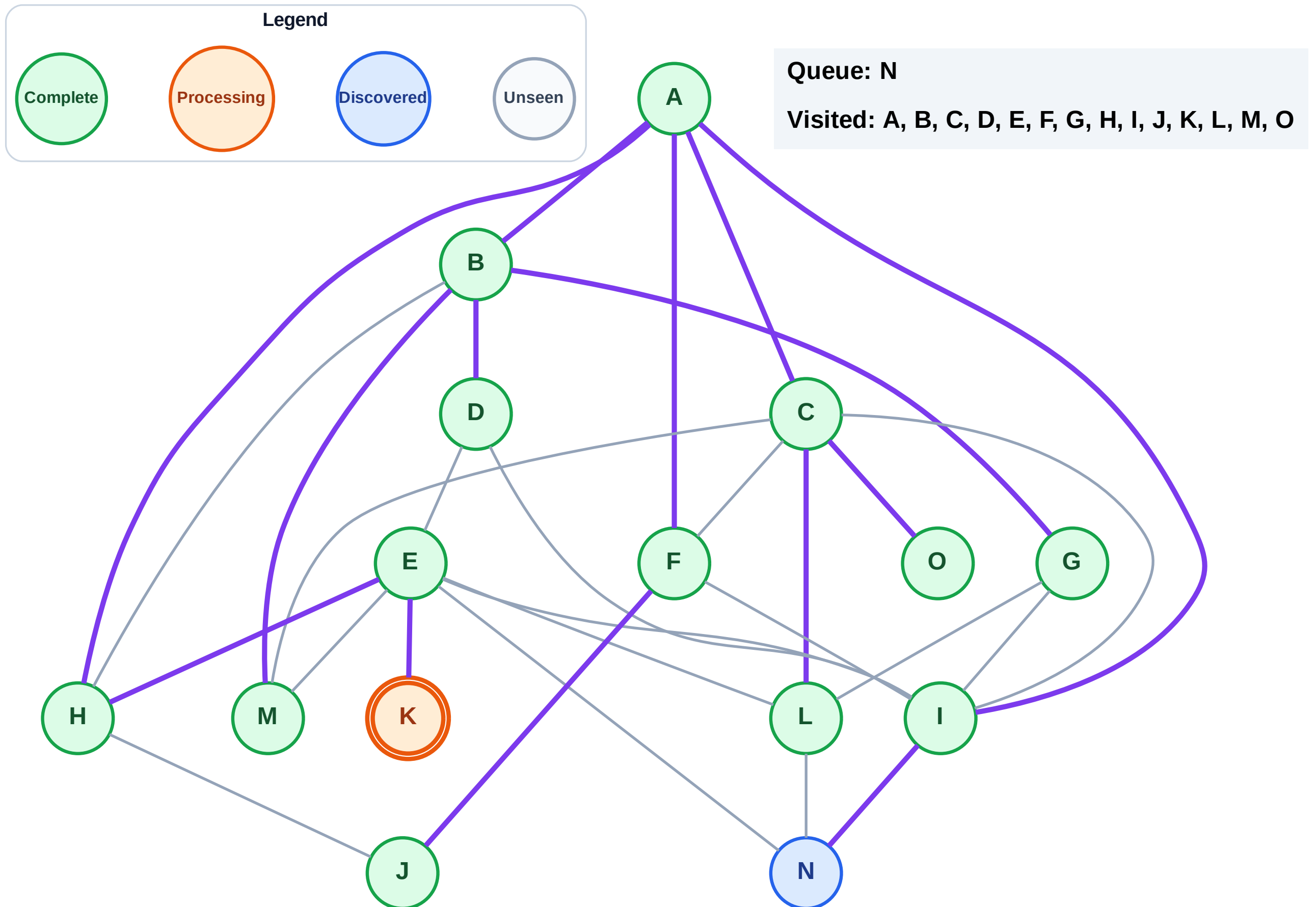
Queue: N

Visited: A, B, C, D, F, G, H, I, J, L, M, O



BFS Traversal

Step 27: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → H → E → K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

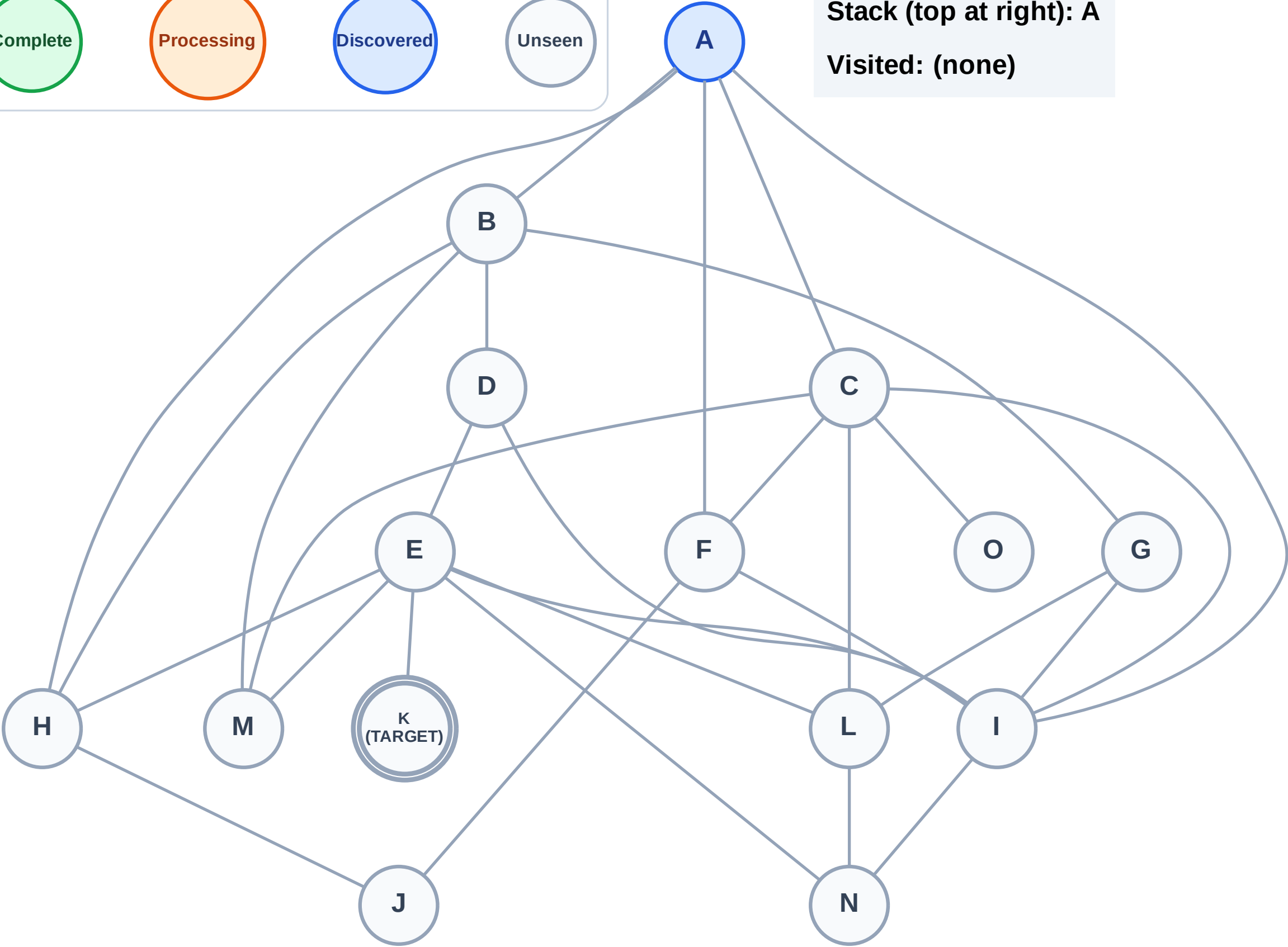
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

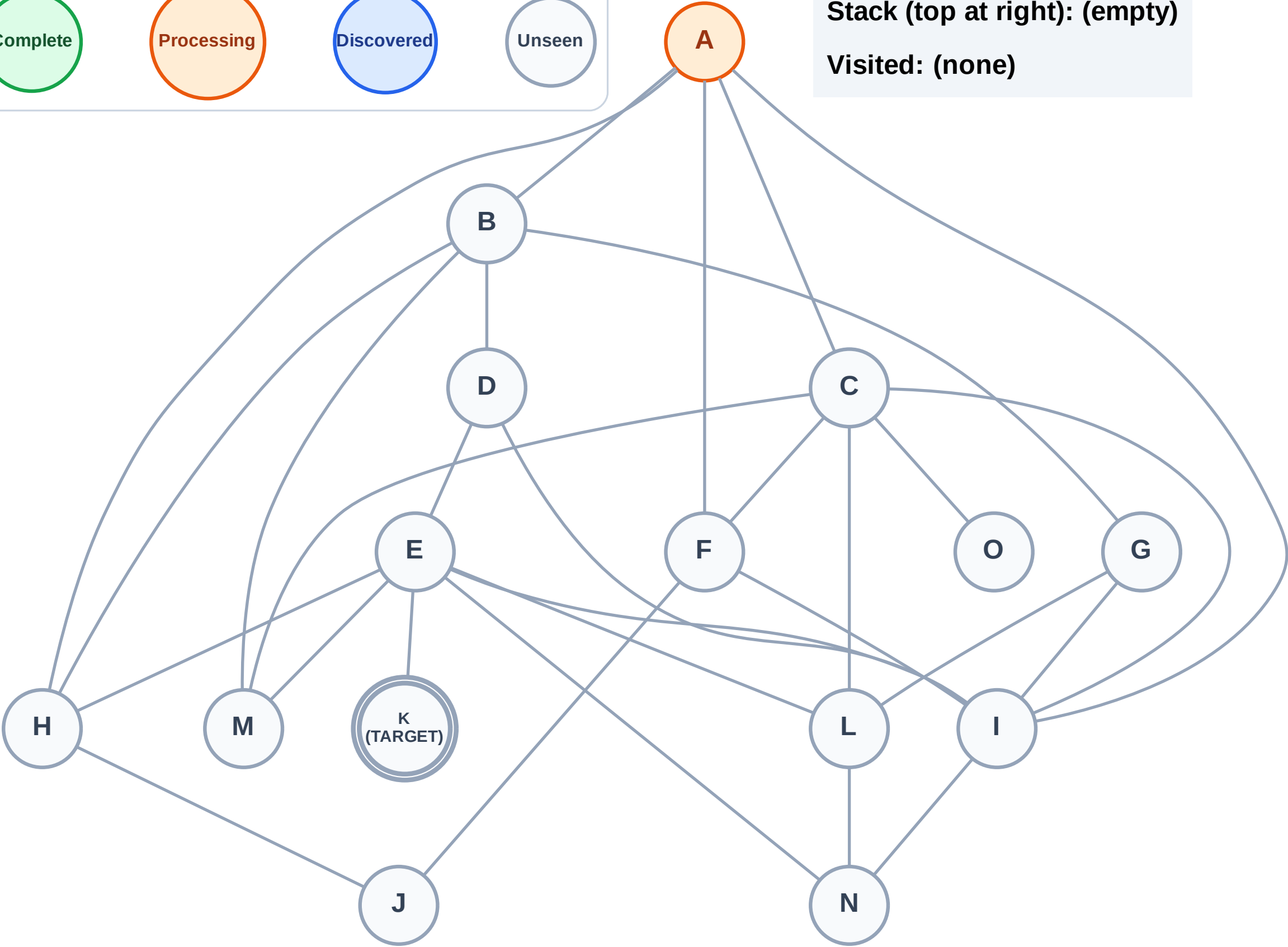
Complete

Processing

Discovered

Unseen

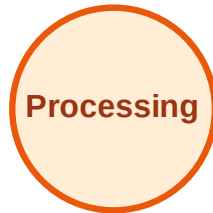
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

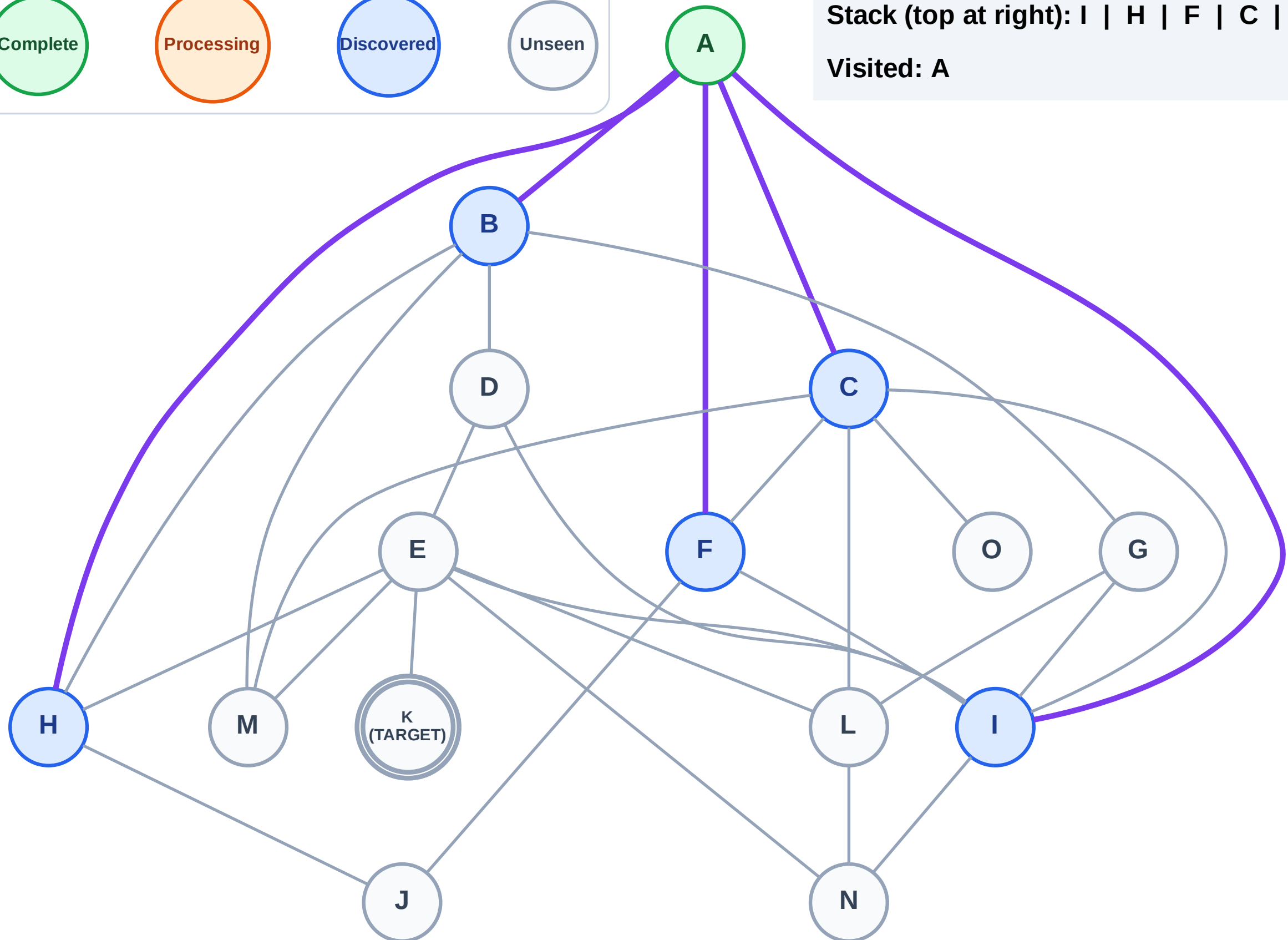
Step 3: Discover I, H, F, C, B from A

Legend



Stack (top at right): I | H | F | C | B

Visited: A

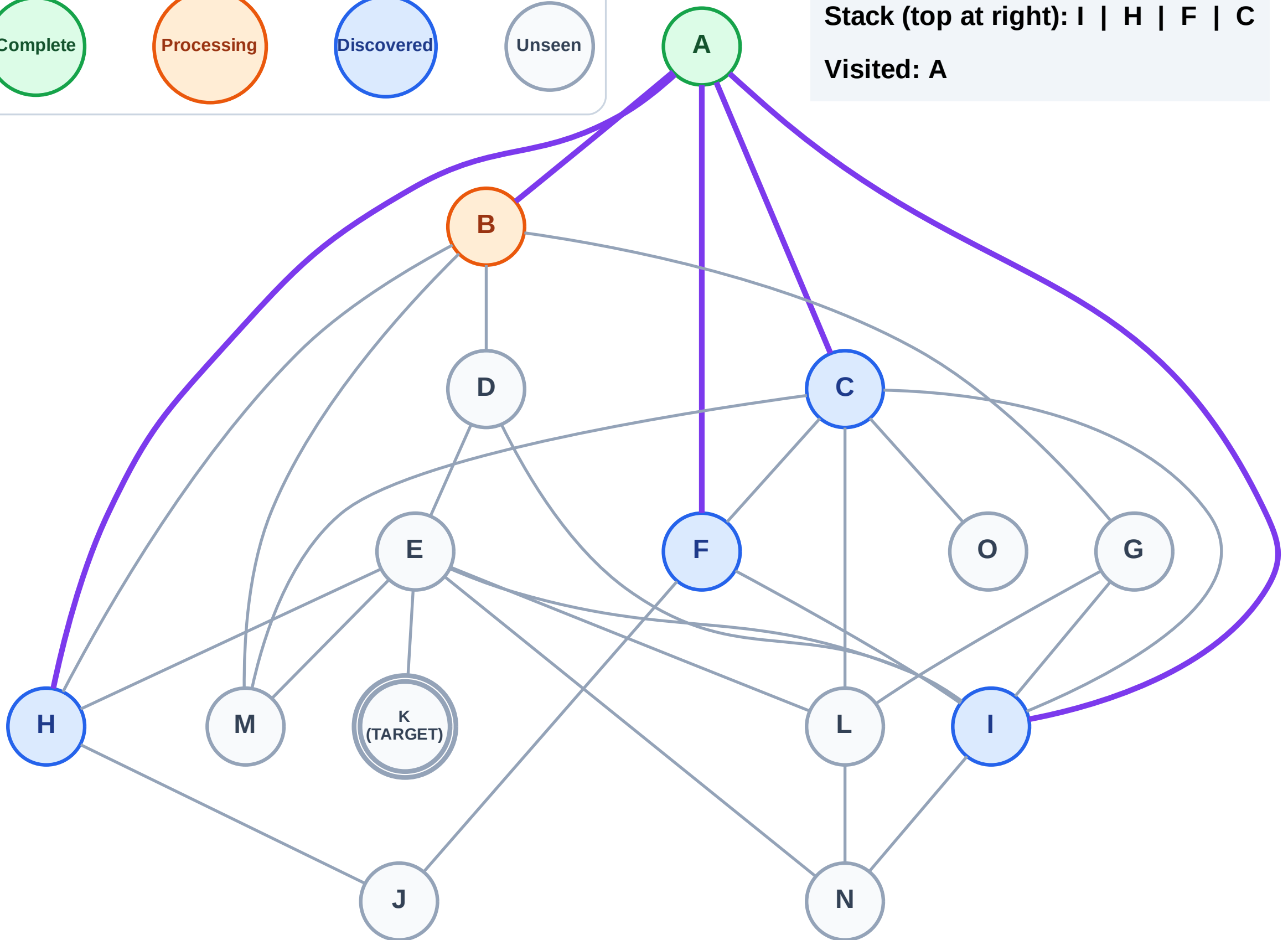


Step 4: Process node B

Legend

Unseen

Visited: A



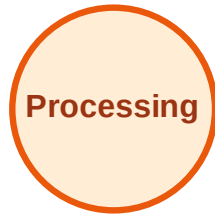
DFS Traversal

Step 5: Discover M, G, D from B

Legend



Complete



Processing



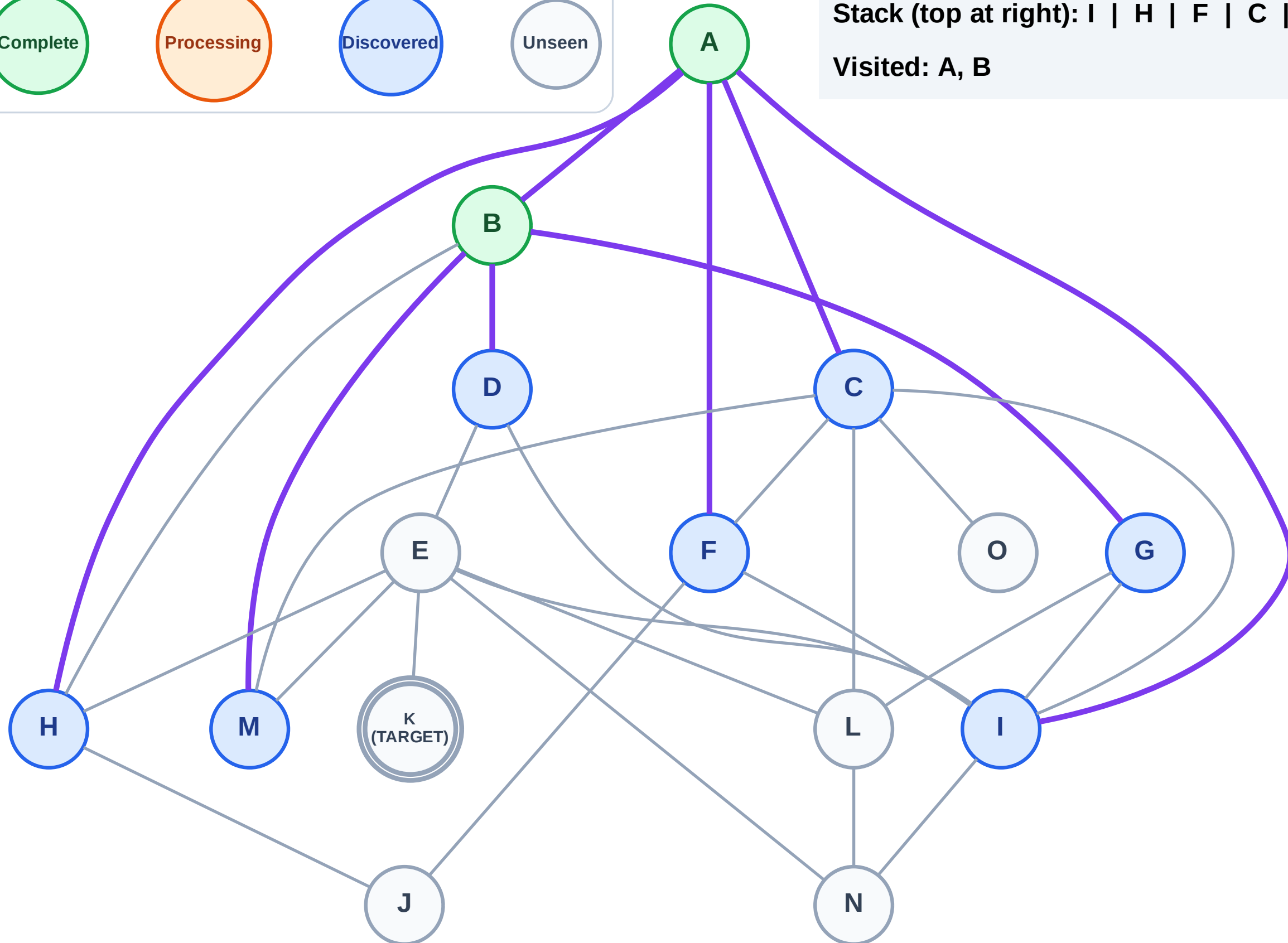
Discovered



Unseen

Stack (top at right): I | H | F | C | M | G | D

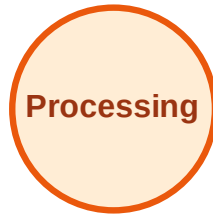
Visited: A, B



DFS Traversal

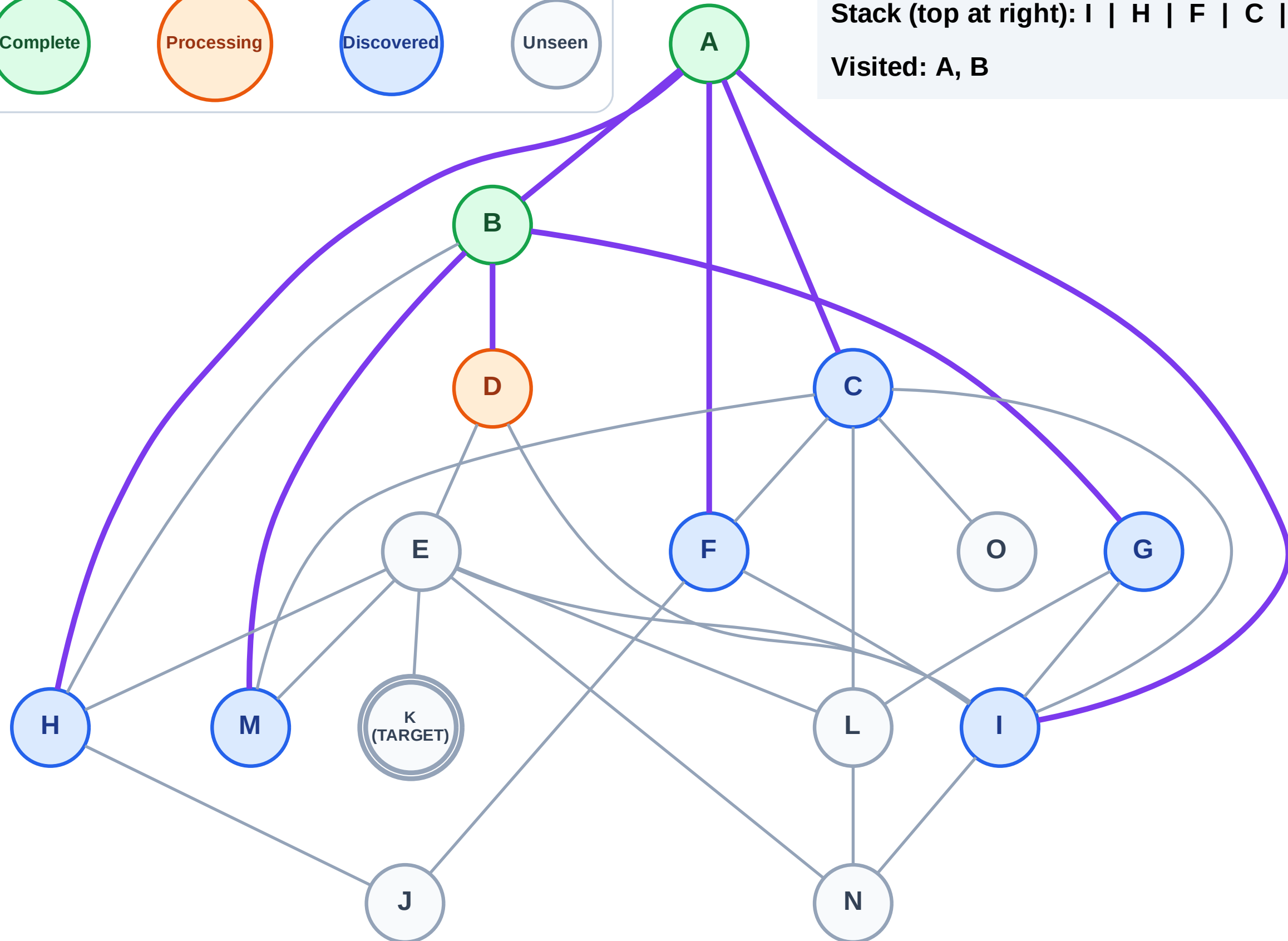
Step 6: Process node D

Legend



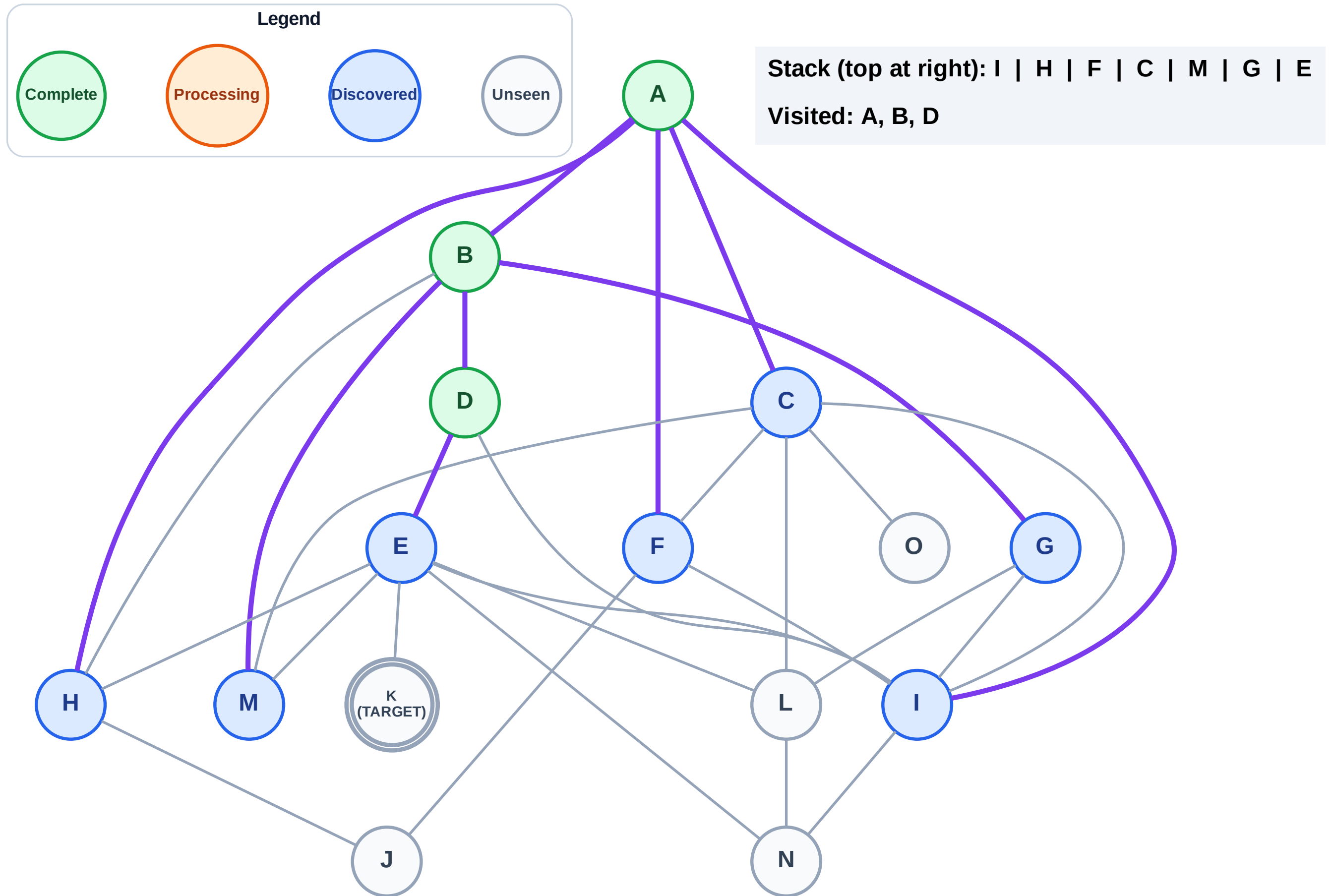
Stack (top at right): I | H | F | C | M | G

Visited: A, B



Step 7: Discover E from D

Step 7: Discover E from D



DFS Traversal

Step 8: Process node E

Legend

Complete

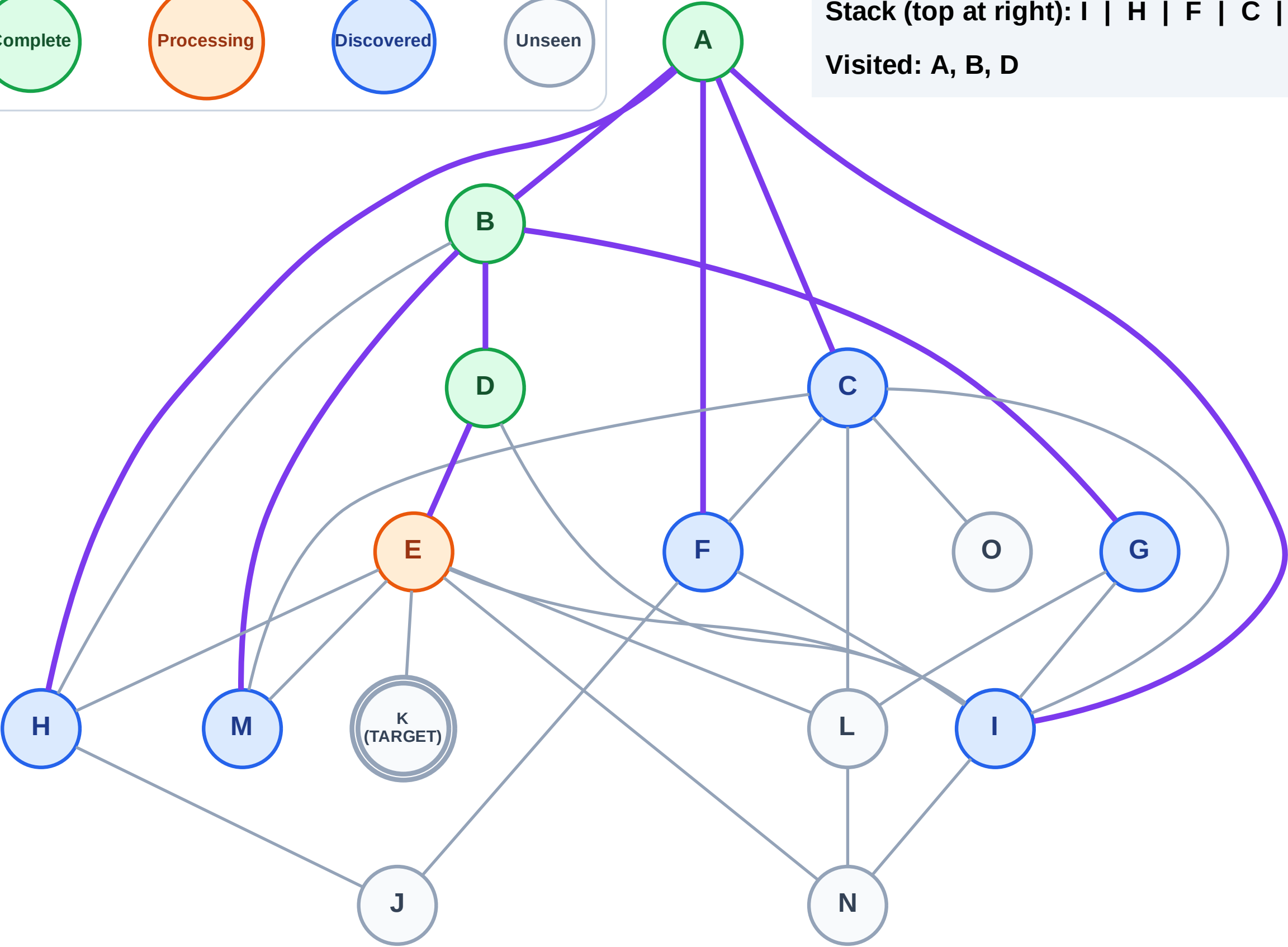
Processing

Discovered

Unseen

Stack (top at right): I | H | F | C | M | G

Visited: A, B, D



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → D → E → K

